STRAIGHT LINES

SYNOPSIS

Inclination of a line:

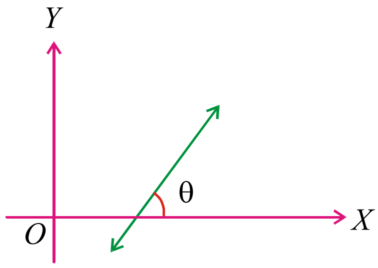
➤ If a line makes an angle $\theta(0 \le \theta < \pi)$ with x-axis measured in positive direction then θ is called inclination of the line.

i) Inclination of horizontal line is zero

ii) Inclination of vertical line is $\pi / 2$

Slope of a line:

➤ If the inclination of a non vertical line is θ then $tan\,\theta$ is called slope of the line and is usually denoted by m, thus $m = tan\,\theta$



i) Slope of horizontal line (x-axis) is zero

$(\because \theta = 0^0)$

ii) Slope of vertical line (y-axis) is not defined $(\because \theta = 90^0)$

iii) $\theta = 0^0 \Leftrightarrow m = 0$

$0^0 < \theta < 90^0 \Leftrightarrow m > 0$

$\theta = 90^0 \Leftrightarrow m$ is not defined

$90^0 < \theta < 180^0 \Leftrightarrow m < 0$

➤ Slope of the line joining two points $A(x_1, y_1)$, $B(x_2, y_2)$ is $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ $(x_1 \ne x_2)$

i) If x_1=x_2 then the line $\overline{AB}$ is vertical and hence its slope is not defined

ii) If y_1=y_2 then the line $\overline{AB}$ is horizontal and hence its slope is 0

➤ Two nonvertical lines are parallel if their slopes are equal.

➤ Two non vertical lines are perpendicular if product of their slopes is -1

W.E-1: ***The medians AD and BE of the triangle with vertices A(0,b), B(0,0) and C(a,0) are mutually perpendicular if***

Sol: $AD \perp BE \Rightarrow \left(\dfrac{-2b}{a}\right)\left(\dfrac{b}{a}\right) = -1$

$\Rightarrow 2b^2 = a^2$

➤ If θ is an angle between two nonvertical lines having slopes m_1, m_2 then

$$\tan\theta = \pm\frac{m_1 - m_2}{1 + m_1 m_2}, m_1 m_2 \ne -1$$

i) If θ is acute then $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right|$

ii) If θ is one angle between two lines then the other angle is $\pi - \theta$. Usually the acute angle between two lines is taken as the angle between the lines

Intercept(s) of a line:

➤ If a line cuts x-axis at A(a, 0) and y-axis at B(0,b) then a and b are called x-intercept and y-intercept of that line respectively

i) Intercept of a line may be positive or negative or zero

ii) x-intercept of a horizontal line is not defined

iii) y-intercept of a vertical line is not defined

iv) Intercepts of a line passing through origin are zero.

Equation of a straight line in various forms:

➤ **i) Line parallel to x-axis:** Equation of horizontal line passing through (a,b) is y = a

ii) Line parallel to y-axis: Equation of vertical line passing through (a,b) is x = b

iii) Slope - point form : The equation of the line with slope **m** and passing through the point (x_1, y_1) is $y - y_1 = m(x - x_1)$

W.E-2: ***If (3,-1),(2,4),(-5,7) are the mid points of the sides $\overline{BC},\overline{CA},\overline{AB}$ of triangle ABC. Then the equation of the side $\overline{CA}$ is***

Sol : Here $m = -1$ and given point (x_1, y_1) is (2, 4). By point slope form equation of the line is

$y - 4 = -1(x - 2)$

iv) Two - point form :

The equation of a line passing through two points $A(x_1,y_1)$ and $B(x_2,y_2)$ is

$$(y-y_1)(x_2-x_1)=(x-x_1)(y_2-y_1)$$

(or) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}=0$

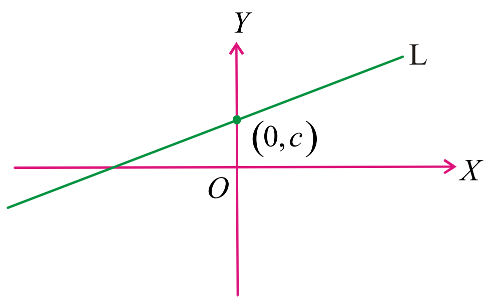
W.E-3: ***Equation of the diagonal (through the origin) of the quadrilateral formed by the lines x = 0, y = 0, x + y = 1 and 6x + y = 3 is***

Sol : Here $(x_1,y_1)=(0,0)$, $(x_2,y_2)=\left(\frac{2}{5},\frac{3}{5}\right)$

Using two-point form, the equation of the line is $3x - 2y = 0$

v) Slope - Intercept form :

a) The equation of the line whose slope is m and which cuts an intercept 'c' on the y-axis is $y = mx+c$



b) The equation of the line whose slope is m and which cuts an intercept 'a' on the x-axis is $y = m(x - a)$

c) The equation of the line passing through the origin and having slope m is $y=mx$

W.E-4: ***Equation to the straight line cutting off an intercept 2 from negative y axis and inclined at 30° to the positive direction of axis of x, is***

Sol : Equation of line passing through (0,-2) and having slope $\frac{1}{\sqrt{3}}$ is $\sqrt{3}y - x + 2\sqrt{3} = 0$

vi) Intercept Form :

Suppose a line L makes intercept on x-axis is a and on y-axis is b then its equation is $\frac{x}{a}+\frac{y}{b}=1$

a) If the portion of the line intercepted between the axes is divided by the point (x_1, y_1) in the ratio m : n, then the equation of the line is $\frac{nx}{x_1}+\frac{my}{y_1}=m+n$

(or) $\frac{mx}{x_1}+\frac{ny}{y_1}=m+n$

b) Equation of the line whose intercept between the axes is bisected at the point (x_1, y_1) is $\frac{x}{x_1}+\frac{y}{y_1}=2$

c) Equation of the line making equal intercepts on the axes and through the point (x_o, y_o) is $x + y = x_o + y_o$

d) Equation of the line making equal intercepts in magnitude but opposite in sign and passing through (x_o,y_o) is $x - y = x_o - y_o$

e) The equation of the line passing through the point (x_1, y_1) and whose intercepts are in the ratio m : n is $nx+my=nx_1+my_1$ (or) $mx+ny=mx_1+ny_1$

W.E-5: ***The sum of x,y intercepts made by the lines $x+y=a$, $x+y=ar$, $x+y=ar^2$ ...... on coordinate axes when r=1/2, $a \neq 0$***

Sol: required sum

$= 2a + 2ar + 2ar^2 + ...(\text{infinite } G.P)$

$= 2a/1-r = 4a$

vii) General equation of line :

a) A linear equation in x and y always represents a line.

b) The equation of a line in general form is $ax+by+c=0$, where a,b,c are real numbers such that $a^2+b^2 \neq 0$ having slope =-a/b ,

x-intercept=-c/a , y- intercept =-c/b .

c) The equation of a line parallel to $ax+by+c=0$ is of the form $ax+by+k=0, k \in R$.

d) The equation of a line perpendicular to $ax+by+c=0$ is of the form $bx-ay+k=0,\ k \in R$

e) Equation of a line passing through (x_1, y_1) and (i) parallel to $ax+by+c=0$ is

$$a(x-x_1)+b(y-y_1)=0$$

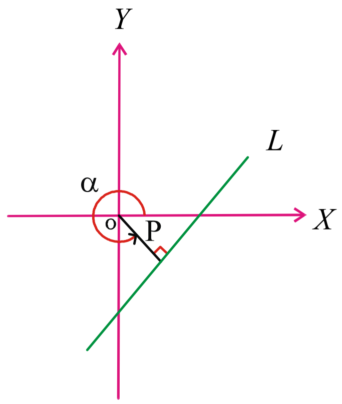
(ii)Perpendicular to $ax+by+c=0$ is

$$b(x-x_1)-a(y-y_1)=0$$

viii) Normal form :

a) The equation of the straight line upon which the length of the normal drawn from origin is 'p' and this perpendicular makes an angle $\alpha, (0 \le \alpha < 2\pi)$ with positive x-axis is

$$x\cos\alpha + y\sin\alpha = p,\ (p>0)$$



b) The normal form of a line $ax+by+c=0$ is

$$\frac{(-a)}{\sqrt{a^2+b^2}}x+\frac{(-b)}{\sqrt{a^2+b^2}}y=\frac{c}{\sqrt{a^2+b^2}}, if\ c>0$$

and

$$\frac{a}{\sqrt{a^2+b^2}}x+\frac{b}{\sqrt{a^2+b^2}}y=\frac{-c}{\sqrt{a^2+b^2}}, if\ c<0$$

W.E-6: ***Normal form of the equation x+y+1=0 is***

Sol: The given equation is x+y+1=0$\Rightarrow$-x-y=1

$$\Rightarrow \frac{(-1)x}{\sqrt{2}}+\frac{(-1)y}{\sqrt{2}}=\frac{1}{\sqrt{2}}$$

$$\Rightarrow x\cos\left(\pi+\frac{\pi}{4}\right)+y\sin\left(\pi+\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$$

$$\Rightarrow x\cos\frac{5\pi}{4}+y\sin\frac{5\pi}{4}=\frac{1}{\sqrt{2}}$$

ix) Symmetric form and Parametric equations of a straight line :

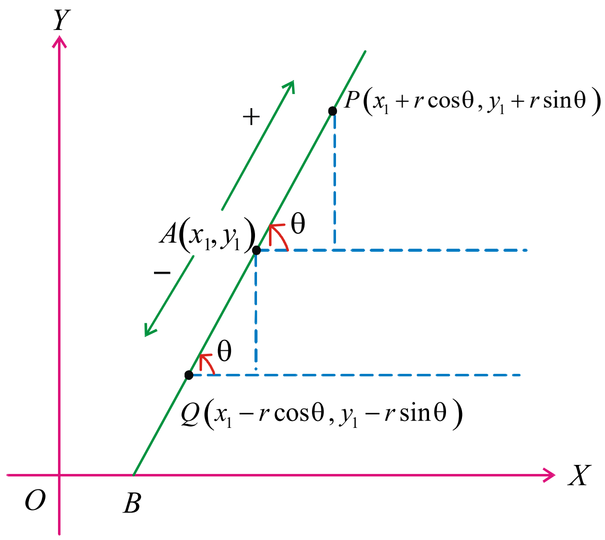
a) The equation of the straight line passing through (x_1,y_1) and makes an angle θ with the positive direction of x-axis is

$$\frac{x-x_1}{\cos\theta}=\frac{y-y_1}{\sin\theta}$$

Where $\theta \in (0,\pi/2) \cup (\pi/2,\pi)$

b) The co-ordinates (x,y) of any point P on the line at a distance 'r' units away from the point $A(x_1,y_1)$ can be taken as $(x_1+r\cos\theta,\ y_1+r\sin\theta)$ (or) $(x_1-r\cos\theta, y_1-r\sin\theta)$

c) The equations $x=x_1+r\cos\theta$, $y=y_1+r\sin\theta$ are called parametric equations of a line with parameter **'r'** of the line passing through the point (x_1,y_1) and having inclination θ.



$\cos\theta = \frac{x - x_1}{AP}$, $\sin\theta = \frac{y - y_1}{AP}$

or $x - x_1 = AP\cos\theta$, $y - y_1 = AP\sin\theta$.

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

W.E-7: ***(1,2),(3,6) are two opposite vertices of a rectangle and if the other two vertices lie on the line 2y = x + c, then c and other two vertices are***

Sol: Mid point of given vertices is

$P(x_1, y_1) = (2,4)$ which lies on $2y = x + c$

then $c=6$.

Now $r=BP=AP=\sqrt{5}$, $\tan\theta = \frac{1}{2}$

Hence B$=(x_1 + r\cos\theta,\ y_1 + r\sin\theta)$=(4,5)

C$=(x_1 - r\cos\theta, y_1 - r\sin\theta)$ =(0,3)

Distances:

➢ i) The perpendicular distance to the line $ax + by + c = 0$

(a) from origin is $\frac{|c|}{\sqrt{a^2 + b^2}}$

(b) from the point (x_1, y_1) *is* $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

ii) The distance of a point (x_1, y_1) from the line $L \equiv ax+by+c=0$ measured along a line making an angle α with x-axis is $\left|\frac{ax_1 + by_1 + c}{a\cos\alpha + b\sin\alpha}\right|$

iii) The distance between parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$.

iv) The distance between the parallel lines $ax+by+c_1=0$ and $ax + by + c_2 = 0$ measured along the line having inclination θ is

$$\left|\frac{c_1 - c_2}{a\cos\theta + b\sin\theta}\right|$$

v) The equation of a line parallel and lying midway between the above two lines is

$$ax + by + \frac{c_1 + c_2}{2} = 0$$

vi) Equiation of the line parallel to ax+by+c=0 and at a distance d from the line is

$$ax + by + c \pm d\sqrt{a^2 + b^2} = 0$$

W.E-8: ***The distance between A(2, 3) on the line of gradient 3/4 and the point of intersection P of this line with 5x + 7y + 40 = 0 is***

Sol : Since m = 3/4, then $\cos\theta$ = 4/5 and $\sin\theta$ = 3/5.

$$r = \left|\frac{5\times2+7\times3+40}{5\left(\frac{4}{5}\right)+7\left(\frac{3}{5}\right)}\right| = \frac{355}{41}$$

Position of a point (s) w.r.to line (s):

➢ i) The ratio in which the line $L \equiv ax + by + c = 0$ divides the line segment joining $A(x_1, y_1)$ *and* $B(x_2, y_2)$ is $-L_{11} : L_{22}$ where

$L_{11} = ax_1 + by_1 + c$, $L_{22} = ax_2 + by_2 + c$

ii) The points A , B lie on the same side or opposite side of the line L = 0 according as L_{11}, L_{22} have same sign or opposite sign that is $L_{11}.L_{22} > 0$ *or* $L_{11}.L_{22} < 0$

W.E-9: ***The range of θ in the interval $(0, \pi)$ such that the points (3, 5) and $(\sin\theta, \cos\theta)$ lie on the same side of the line x + y – 1 = 0 is***

Sol : Since $(3+5-1)(\sin\theta + \cos\theta - 1) > 0$

$$\Rightarrow \quad \sin\left(\frac{\pi}{4} + \theta\right) > \frac{1}{\sqrt{2}}$$

$$\Rightarrow \quad \frac{\pi}{4} < \frac{\pi}{4} + \theta < \frac{3\pi}{4}$$

$$\Rightarrow \quad 0 < \theta < \frac{\pi}{2}$$

iii) A point $A(x_1, y_1)$ and origin lies on the same or opposite side of a line $L = ax + by + c = 0$ according as $c.L_{11} > 0$ *or* $c.L_{11} < 0$

iv) The point (x_1, y_1) lies between the parallel lines $ax_1 + by_1 + c = 0$, $ax_2 + by_2 + c = 0$ or does not lie between them according as $\frac{ax_1+by_1+c_1}{ax_1+by_1+c_2}$ is negative or positive

v) The point $A(x_1,y_1)$ lies above or below the line $L = ax+by+c=0$ according as $\frac{L_{11}}{b}>0 \ or \ \frac{L_{11}}{b}<0$

Proof: The fig. Shows a point $P(x_1, y_1)$ lying above a given line. If an ordinate is dropped from P to meet the line L at N, then the x coordinate of N will be x_1.

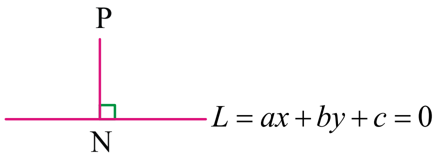
Putting $x = x_1$ in the equation $ax + by + c = 0$

gives ordinate of N $= -\frac{(ax_1+c)}{b}$

If $P(x_1, y_1)$ lies above the line, then we have

$y_1 > -\frac{(ax_1+c)}{b}$ i.e. $y_1+\frac{(ax_1+c)}{b}>0$

i.e. $\frac{(ax_1+by_1+c)}{b}>0$, i.e. $\frac{L(x_1,y_1)}{b}>0$



Hence, $P(x_1, y_1)$ lies above the line $ax + by + c = 0$, and if $\frac{L(x_1,y_1)}{b}<0$, it would mean that P lies below the line $ax + by + c = 0$.

➢ **If $P(x_1,y_1)$ lie between the parallel lines $ax+by+c_1=0, ax+by+c_2=0$ then $(ax_1+by_1+c_1)(ax_1+by_1+c_2)<0$.**

➢ **If $P(x_1,y_1)$ does not lie between the parallel lines** $ax+by+c_1=0, ax+by+c_2=0$ then $(ax_1+by_1+c_1)(ax_1+by_1+c_2)>0$

Proof :

Make c_1,c_2 having same sign.

(If necessary)

$\Rightarrow$ (0,0) lie on same side of both the lines

$\Rightarrow ax_1+b_1y_1+c_1,\ c_1$ have opposite signs

$ax_1+b_1y_1+c_2,\ c_2$ have opposite signs

since $c_1c_2>0$, we have

$(ax_1+by_1+c_1)(ax_1+by_1+c_2)>0$

Ceva's Theorem :

➢ If the lines joining any point 'O' to the vertices A,B,C of a triangle meet the opposite sides in D,E, F respectively then $\frac{BD}{DC}.\frac{CE}{EA}.\frac{AF}{FB}=1$

Proof: Without loss of generality take the point P as the origin O.

Let $A(x_1,y_1), B(x_2,y_2), C(x_3,y_3)$ be the vertices. Slope of $\overline{AP}$ is $\frac{y_1-0}{x_1-0}=\frac{y_1}{x_1}$

Equation of $\overline{AP}$ is $y-0=\frac{y_1}{x_1}(x-0)$

$\Rightarrow yx_1 = xy_1 \Rightarrow xy_1 - yx_1 = 0$

$\therefore \frac{BD}{DC}=\frac{-(x_2y_1-x_1y_2)}{x_3y_1-x_1y_3}=\frac{x_1y_2-x_2y_1}{x_3y_1-x_1y_3}$

$\therefore \frac{BD}{DC}.\frac{CE}{EA}.\frac{AF}{FB}=1$

Manelaw's Theorem :

➢ If a transversal cuts the sides BC, CA, AB of a triangle in D, E, F respectively then $\frac{BD}{DC}.\frac{CE}{EA}.\frac{AF}{FB}=-1$

Proof: Let the transversal be ax+by+c=0. the line divides $\overline{BC}$ at D then

$\frac{BD}{DC}=\frac{-(ax_2+by_2+c)}{(ax_3+by_3+c)}$

Hence $\frac{BD}{DC}.\frac{CE}{EA}.\frac{AF}{FB}=-1$

W.E-10: ***The range of α, if (α, α^2) lies inside the triangle having sides along the lines*** $2x + 3y = 1,\ x + 2y - 3 = 0,\ 6y = 5x - 1$

Sol : Let A, B, C be vertices of the triangle.

$A = (-7, 5),\ B = \left(\frac{5}{4}, \frac{7}{8}\right)$

$C = \left(\frac{1}{3}, \frac{1}{9}\right)$. Sign of A w.r.t. BC to -ve.

If P lies inside the triangle ABC, then sign of P will be the same as sign of A w.r.t. the line BC

$\Rightarrow \quad 5\alpha - 6\alpha^2 - 1 < 0$ .....(i)

similarly $2\alpha + 3\alpha^2 - 1 > 0$ .....(ii)

And $\quad \alpha + 2\alpha^2 - 3 < 0$ .....(iii)

Solving (i), (ii) and (iii) for α and then taking intersection,

we get $\alpha \in \left(\frac{1}{2}, 1\right) \cup \left(-\frac{3}{2}, -1\right)$

Point of intersection of lines and Concurrency of Straight Lines:

➢ i) Consider two lines $L_1 \equiv a_1x + b_1y + c_1 = 0$ and $L_2 \equiv a_2x + b_2y + c_2 = 0$ then point of intersection is

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right) or$$

$$\frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

ii) Three or more lines are said to be concurrent, if they have a point in common. The common point is called the point of concurrence.

a) If $L_1 = 0,\ L_2 = 0$ are two interesecting lines, then the equation of any line other than $L_1 = 0$ and $L_2 = 0$ passing through point of intersection can be taken as $L_1 + \lambda L_2 = 0$. Where λ is a parameter

b) The three lines $L_i \equiv a_ix + b_iy + c_i = 0,\ i = 1,2,3$ are concurrent *iff* $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

(or) Point of intersection of any two lines lies on the third line

(or) there exist constants $\lambda_1, \lambda_2, \lambda_3$ not all zero such that $\lambda_1L_1 + \lambda_2L_2 + \lambda_3L_3 = 0$

c) If $p_1x+q_1y = 1$, $p_2x+q_2y = 1$, $p_3x+q_3y = 1$ are concurrent lines then the points (p_1,q_1), (p_2,q_2), (p_3,q_3) are collinear

d) If $ka+lb+mc=0$, then the point of concurrency of the lines represented by $ax+by+c=0$ is $\left(\frac{k}{m}, \frac{l}{m}\right)$

W.E-11: ***The line $x + \lambda y - 4 = 0$ passes through the point of intersection of $4x - y + 1 = 0$ and $x + y + 1 = 0$. Then the value of λ is***

Sol : The three lines are concurrent

$$\Rightarrow \begin{vmatrix} 1 & \lambda & -4 \\ 4 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$\Rightarrow \quad -2 - 3\lambda - 20 = 0 \Rightarrow \quad \lambda = -\frac{22}{3}$

Angle between lines:

➢ i) If 'θ' is an acute angle between the lines having slopes m_1 *and* m_2 then

$$tan\,\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right|$$

ii) If 'θ' is an acute angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ then

$$\cos\theta = \left|\frac{a_1a_2 + b_1b_2}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}\right| \ and \ \ tan\theta = \left|\frac{a_1b_2 - a_2b_1}{a_1a_2 + b_1b_2}\right|$$

other angle between the lines is $\pi - \theta$

iii) The slope m of a line which is equally inclined with two intersecting lines of slopes m_1 *and* m_2 is given by

$\frac{m_1 - m}{1 + mm_1} = \frac{m - m_2}{1 + m m_2}$

iv) The slopes of the lines making an angle α with a line having slope m are $\frac{m - \tan\alpha}{1 + m\tan\alpha}, \frac{m + \tan\alpha}{1 - m\tan\alpha}$

v) Consider two lines $L_1 = a_1x + b_1y + c_1 = 0$ and $L_2 = a_2x + b_2y + c_2 = 0$

a) Lines are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

b) Lines are coincident if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

c) Lines are perpendicular if $a_1a_2 + b_1b_2 = 0$

d) Lines are equally inclined with x-axis

$if\ \frac{a_1}{a_2} = -\frac{b_1}{b_2}$

W.E-12: ***A straight line through (2, 2) intersects the lines $\sqrt{3}x + y = 0$ and $\sqrt{3}x - y = 0$ at the points A and B. The equation to the line AB so that the triangle OAB is equilateral is***

Sol: Since given two lines passing through origin and making angles 60^0, 120^0 with X-axis the third line is parallel to X-axis. Hence equation of AB is y=2

Triangles and Quadrilaterals:

➤ i) The ratio of the sides of a triangle formed by $L_1 = 0,\ L_2 = 0\ and\ L_3 = 0$ is

$$\sqrt{a_1^2+b_1^2}\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} : \sqrt{a_2^2+b_2^2}\begin{vmatrix} a_3 & b_3 \\ a_1 & b_1 \end{vmatrix} : \sqrt{a_3^2+b_3^2}\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

ii) Let d_1 be the distance between the parallel lines $ax + by + c_1 = 0$, $ax + by + c_2 = 0$ and d_2 be the distance between the parallel lines $a_1x + b_1y + k_1 = 0$, $a_1x + b_1y + k_2 = 0$ then the figure formed by four lines is

a) a square if $d_1 = d_2$ and $aa_1 + bb_1 = 0$,

b) Rhombus if $d_1 = d_2$ and $aa_1 + bb_1 \neq 0$,

c) Rectangle if $d_1 \neq d_2$ and $aa_1 + bb_1 = 0$,

d) Parallelogram if $d_1 \neq d_2$ and $aa_1 + bb_1 \neq 0$

➤ i)The area of triangle formed by the line $\frac{x}{a} + \frac{y}{b} = 1$ with the co- ordinate axis is $\frac{1}{2}|ab|$

ii)The area of triangle formed by line $ax + by + c = 0$ with the co - ordinate axes is $\frac{c^2}{2|ab|}$

iii) Area of the rhombus $a|x| + b|y| + c = 0$ is $4(area\ of\ \Delta) = \frac{2c^2}{|ab|}$

iv) The area of triangle formed by lines $a_ix + b_iy + c_i = 0,\ i = 1,2,3$ is $= \frac{\Delta^2}{2|\lambda_1\lambda_2\lambda_3|}$

$$where\ \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \lambda_1 = \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix},$$

$$\lambda_2 = \begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix}, \lambda_3 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

v) The area of triangle formed by lines

$y = m_ix + c_i,\ i = 1,2,3\ is\ \frac{1}{2}\left|\sum \frac{(c_1 - c_2)^2}{m_1 - m_2}\right|$

vi) If p_1, p_2 are distances between parallel sides and 'θ' is angle between adjacent sides of parallelogram then its area is $\left|\frac{p_1p_2}{\sin\theta}\right|$

vii) Area of parallelogram whose sides $a_1x + b_1y + c_1 = 0$, $a_1x + b_1y + c_2 = 0$, $a_2x + b_2y + d_1 = 0$ and $a_2x + b_2y + d_2 = 0$ is $\left|\frac{(c_1 - c_2)(d_1 - d_2)}{a_1b_2 - a_2b_1}\right|$

viii) Area of rhombus $= \frac{1}{2}\ d_1d_2$ where d_1, d_2 are lengths of the diagonals

W.E-13: *The triangle formed by the lines* ***x-7y-22=0, 3x+4y+9=0, 7x+y-54=0 is***

So: by using

$$\sqrt{a_1^2+b_1^2}\begin{vmatrix}a_2 & b_2\\ a_3 & b_3\end{vmatrix}:\sqrt{a_2^2+b_2^2}\begin{vmatrix}a_3 & b_3\\ a_1 & b_1\end{vmatrix}:\sqrt{a_3^2+b_3^2}\begin{vmatrix}a_1 & b_1\\ a_2 & b_2\end{vmatrix}$$

we get $1:\sqrt{2}:1$ hence the triangle is right angled isosceles.

W.E-14: ***If the distance of any point P(x, y) from the origin is defined as d(x, y) = Max. (|x|, |y|) and d(x, y) = a (non zero constant), then the locus of the P is***

Sol: d(x, y) = Max. (|x|, |y|) .....(i)

But d(x, y) = a ..... (ii)

From (i) and (ii),

a = Max. (|x|, |y|)

If |x| > |y|, then a = |x| $\Rightarrow$ x = ± a

If |y| > |x|, then a = |y| $\Rightarrow$ y = ± a

Hence locus of P represents a square.

Foot and Image:

➢ i) If (h,k) is the foot of the perpendicular from (x_1,y_1) to the line $ax+by+c=0$ then

$$\frac{h-x_1}{a}=\frac{k-y_1}{b}=\frac{-(ax_1+by_1+c)}{a^2+b^2} \text{ or}$$

$(h,k)=(x_1+a\lambda, y_1+b\lambda)$ where

$$\lambda=\frac{-(ax_1+by_1+c)}{a^2+b^2}$$

ii) If (h,k) is the image (reflection) of the point (x_1,y_1) w.r.t the line $ax+by+c=0$ then

$$\frac{h-x_1}{a}=\frac{k-y_1}{b}=\frac{-2(ax_1+by_1+c)}{a^2+b^2} \text{ or}$$

$(h,k)=(x_1+a\lambda, y_1+b\lambda)$ where

$$\lambda=\frac{-2(ax_1+by_1+c)}{a^2+b^2}$$

iii) Image of (a , b) w.r.to y = x is (b , a)

iv) Image of (a , b) w.r.to x + y = 0 is $(-b,-a)$

v) If B is image of A w.r.to P then 2P = A + B

vi) Reflection of $f(x,y)=0$ in x-axis is

$f(x,-y)=0$

vii) Reflection of $f(x,y)=0$ in y-axis is

$f(-x,y)=0$

viii) Reflection of $f(x,y)=0$ in x = y is

$f(y,x)=0$

ix) Image of the line ax+by+c = 0 w.r.t line lx+my+n =0 (or) the straight line lx+my+n=0 bisects an angle between the two lines of which one of them is ax+by+c=0 then equation of other line is

$(l^2+m^2)(ax+by+c)=2(al+bm)(lx+my+n)$

W.E-15 : ***In ΔABC A is (1,2) if the internal angle bisector of B is 2x-y+10=0 and perpendicular bisector of AC is y=x then the equation of BC is***

Sol: Image of A w.r.to bisector of B is (-7,6) lies on BC and image of A in the perpendicular bisector of AC is C(2,1)

$\therefore$ equation of BC is 5x+9y-19=0

Centroid, circumcentre, orthocentre and incentre:

➢ i) Let $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$ be vertices of ΔABC then,

a) Equation of altitude through A is

$(x-x_1)(x_2-x_3)=(y-y_1)(y_2-y_3)$

b) Equation of perpendicular bisector of the side $\overline{AB}$ is

c) Orthocentre of $\Delta\, ABC$ is

$$\left(\frac{x_1\tan A+x_2\tan B+x_3\tan C}{\tan A+\tan B+\tan C},\frac{y_1\tan A+y_2\tan B+y_3\tan C}{\tan A+\tan B+\tan C}\right)$$

d) Circum centre of ΔABC is

$$\left(\frac{x_1\sin 2A+x_2\sin 2B+x_3\sin 2C}{\sin 2A+\sin 2B+\sin 2C},\frac{y_1\sin 2A+y_2\sin 2B+y_3\sin 2C}{\sin 2A+\sin 2B+\sin 2C}\right)$$

ii) The equations of the sides BC,CAandAB of the triangle ABC formed by the lines

$L_i = a_i x + b_i y + c_i = 0$ (i=1,2,3) then

a) Orthocentre is point of intersection of

$(a_2a_3+b_2b_3)L_1=(a_3a_1+b_3b_1)L_2=(a_1a_2+b_1b_2)L_3$

b) Median through A is $\lambda_2 L_2 - \lambda_3 L_3 = 0$

Hence centroid satisfies $\lambda_2 L_2 = \lambda_3 L_3 = \lambda_1 L_1$

iii) If H is orthocentre of triangle ABC, then orthocentre of triangle formed by any three of the points H,A,B,C will be the remaining point.

iv) Circumcentre is equidistant from the vertices of triangle

v) If G is the centroid, H is the orthocentre and S is the circumcentre then

a) The relation between them is 3G = 2S + H.

b) $H = 3G$ when $S = (0,0)$

vi) Incentre is equidistant from all sides of the traingle.

vii) In a triangle ABC,

a) The internal bisector of angle A, ie. AD divides opposite side BC at D in the ratio AB:AC

b) The external bisector of angle A, ie. AD divides opposite side BC at D in the ratio -AB:AC

viii) If the algebraic sum of the perpendicular distances from three points to a variable line is zero, then the line passes through the centroid of the triangle formed by the three points.

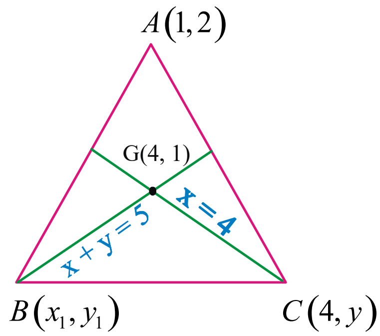
W.E-16: ***In a triangle ABC, coordinates of A are (1, 2) and the equations to the medians through B and C are x + y = 5 and x = 4 respectively. Then the points B and C are***

Sol: Let B be (x_1, y_1) and C be (4, y). Since medians through B and C meet at Centroid G (4, 1)

$$\Rightarrow \frac{x_1+4+1}{3} = 4 \Rightarrow x_1 = 7$$

Since $B(x_1, y_1)$ lies on $x + y = 5$

$\therefore$ $y_1 = 5 - x_1 = 5 - 7 = -2$ $\Rightarrow$ B is (7,–2),



Also $\dfrac{y_1+y+2}{3} = 1 \Rightarrow y = 3 - 2 - y_1$

$\therefore$ C is (4, 3)

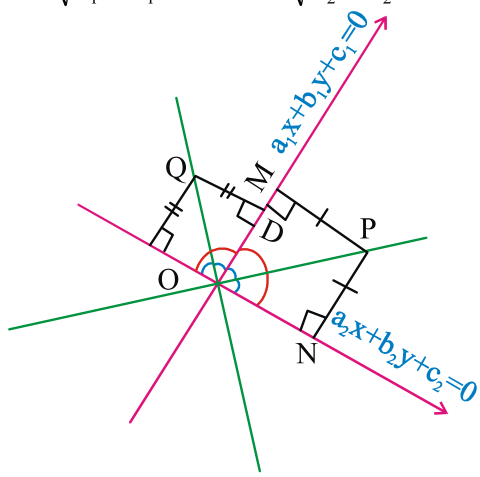
Angular bisectors of two straight lines:

➤ Angular bisector is the locus of a point which moves in such a way so that its distance from two intersecting lines remains same.

The equations of the two bisectors of the angles between the lines $a_1x + b_1y + c_1 = 0$ and

$a_2x + b_2y + c_2 = 0$ are

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = \pm\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$



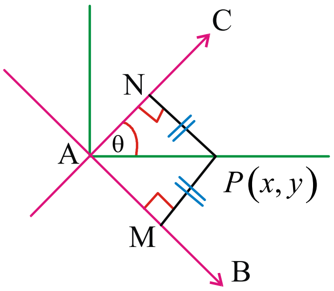
i) If the two given lines are not perpendicular i.e. $a_1 a_2 + b_1 b_2 \neq 0$ and not parallel i.e. $a_1 b_2 \neq a_2 b_1$ then one of these equations is the equation of the bisector of the acute angle between two given lines and the other that of the obtuse angle between two given lines.

ii) Whether both given lines are perpendicular or not, but the angular bisectors of these lines will always be mutually perpendicular.

iii) The bisectors of the acute and the obtuse angles:

Take one of the lines and let its slope be m_1 and take one of the bisectors and let its slope be m_2. If θ be the acute angle between them, then find

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$



If $\tan\theta > 1$ then the bisector taken is the bisector of the obtuse angle and the other one will be the bisector of the acute angle.

If $0 < \tan\theta < 1$ then the bisector taken is the bisector of the acute angle and the other one will be the bisector of the obtuse angles.

iv) consider the lines are $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$, where $c_1 > 0,\ c_2 > 0$ then,

$$\frac{a_1 x + b_1 y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2 x + b_2 y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

will represent the equation of the bisector of the acute or obtuse angle between the lines according as $a_1 a_2 + b_1 b_2$ is negative or positive.

v) The equation of the bisector of the angle which contains a given point :

The equation of the bisector of the angle between the two lines containing the point

(x_1, y_1) is $\dfrac{a_1 x + b_1 y + c_1}{\sqrt{a_1^2 + b_1^2}} = \dfrac{a_2 x + b_2 y + c_2}{\sqrt{a_2^2 + b_2^2}}$

or $\dfrac{a_1 x + b_1 y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\dfrac{a_2 x + b_2 y + c_2}{\sqrt{a_2^2 + b_2^2}}$

according as $a_1 x_1 + b_1 y_1 + c_1$ and $a_2 x_1 + b_2 y_1 + c_2$ are of the same signs or of opposite signs.

vi) For example the equation of the bisector of the angle containing the origin is given by

$\dfrac{a_1 x + b_1 y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \dfrac{a_2 x + b_2 y + c_2}{\sqrt{a_2^2 + b_2^2}}$ for same sign of c_1 and c_2 (for opposite sign take –ve sign in place of +ve sign)

vii) If $c_1 c_2 (a_1 a_2 + b_1 b_2) < 0$, then the origin will lie in the acute angle and if $c_1 c_2 (a_1 a_2 + b_1 b_2) > 0$, then origin will lie in the obtuse angle.

viii) Equation of straight lines passing through $P(x_1, y_1)$ and equally inclined with the lines $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$ are those which are parallel to the bisectors between these two lines and passing through the point P.

W.E-17 :

For the straight lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$, find the equation of the -

(i) Bisector of the obtuse angle between them is

ii) Bisector of the acute angle between them

is

(iii) Bisector of the angle which contains origin is

(iv) Bisector of the angle which contains (1, 2) is

Sol: after making $c_1>0$ and $c_2>0$;

$a_1a_2+b_1b_2 = (-4)(5)+(-3)(12) = -56<0$

i) The bisector of the acute angle is

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = \frac{5x+12y+9}{\sqrt{5^2+12^2}}$$

$7x+9y-3=0$

ii) The bisector of the obtuse angle is

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = -\frac{5x+12y+9}{\sqrt{5^2+12^2}}$$

$9x-7y-41=0$

(iii) The bisector of the angle containing the origin

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = \frac{5x+12y+9}{\sqrt{5^2+12^2}}$$

$7x+9y-3=0$

(iv) For the point (1, 2),

$4x+3y-6 = 4\times 1+3\times 2-6>0$

$5x+12y+9 = 12\times 2+9>0$

Hence equation of the bisector of the angle containing the point (1, 2) is

$$\frac{4x+3y-6}{5} = \frac{5x+12y+9}{13}$$

$9x-7y-41=0$

Optimization:

➤ Let A and B are two points on same side of line $L \equiv ax+by+c=0$

i) The point **P** such that PA + PB is minimum, is intersection of $L=0$ and the line joining A to image of B

or line joining B to image of A w.r.to $L=0$

ii) The point is P such that $|PA-PB|$ is Maximum, is point of intersection of line $L=0$ and line joining A and B.

W.E-18: ***A light ray emerging from the point source placed at P(2, 3) is reflected at a point 'Q' on the y-axis and then passes through the point R(5, 10). Coordinate of 'Q' is -***

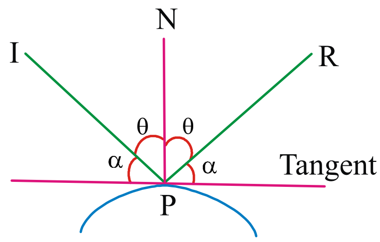
Sol: Image of point P(2,3) in Y-axis is $P^1(-2,3)$

Equation of $P^1R \equiv y-3=1(x+2)$

$x-y+5=0$

P^1R meets the Y-axis at Q(0,5)

Reflection in surface:



IP = incident ray

PN = normal to the surface

PR = reflected ray

$\angle$ IPN = $\angle$NPR

$\therefore$ Angle of incident = Angle of reflection

No. of lines, no. of triangles and no. of circles:

➤ No. of lines drawn through the point A which are at a distance d from the point B

a) If AB = d then the no. of lines through A at a distance d from B is 1

b) If AB>d then the no. of lines through A at a distance d from B is 2

c) If AB<d then the no. of lines through A at a distance d from B is 0

➤ No of right angled traingles in a circle depends on height h of the traingle and radius r of the circle

a) If h = r, no. of right angled traingles = 2

b) If h < r, no. of right angled traingles = 4

c) If h > r, no. of right angled traingles = 0

➢ No. of circles touching three lines

a) No circle if the lines are parallel

b) one circle if the lines are concurrent

c) 2 circles if two lines are parallel and third cuts them

d) 4 circles if the lines are not concurrent and no two of them are parallel.

W.E-19: ***Let A= (1,2), B= (3,4) and C=(x,y) be a point such that(x–1) (x–3)+(y–2)(y–4)=0. If area of Δ ABC=1 then maximum number of positions of C in the xy plane is***

Sol: Ends of diameter are A= (1,2), B= (3,4)

Area of the triangle is equal to 1

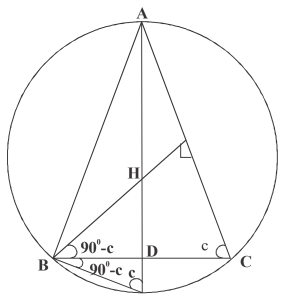
$$\Rightarrow \frac{1}{2}\left(2\sqrt{2}\right)(h)=1 \Rightarrow h=\frac{1}{\sqrt{2}}$$

$$\text{radius}=\frac{AB}{2}=\frac{\sqrt{5}}{2}$$

$\therefore$ number of traingles= 4 ($\because$ h < r)

➢ **Image of orthocentre of $\Delta\, ABC$ w.r.t. a side of the triangle lies on circumcircle of $\Delta\, ABC$**

Proof : From diagram, $\Delta\, BHD$, $\Delta\, BTD$ are congruent triangles $\Rightarrow$ T is image of H w.r.t. $\overline{BC}$

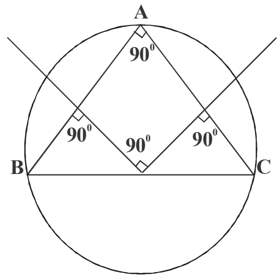


➢ **If the perpendicular bisectors of sides $\overline{AB}$, $\overline{AC}$ are perpendicular then**

i) $\angle BAC$ is 90^0

ii) point of intersection of the perpendicular bisectors is mid point of $\overline{BC}$

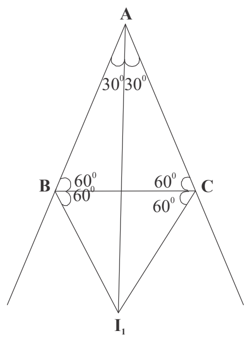
Proof :



➢ **For an equilateral $\Delta\, ABC$, ex-centres I_1, I_2 and I_3 are images of A,B and C w.r.t. $\overline{BC}, \overline{CA}$ & $\overline{AB}$ respectively**

Proof : ΔBAC, ΔBI_1C are similar triangles.

$\Rightarrow I_1$ is image of A w.r.t. $\overline{BC}$

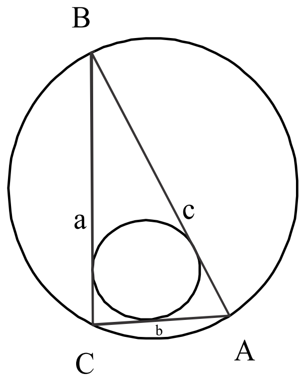


➢ **In a right angled triangle, the sum of the lengths of the legs is equal to the sum of the diameters of the inscribed and the circumscribed circles.**

Proof : c =2R (R is circum radius)

In-radius = r = (s-c) tan C/2

$$\Rightarrow 2r = a + b - c \Rightarrow 2R + 2r = a + b$$



➢ If the sides $\overline{BC}$, $\overline{CA}$ and $\overline{AB}$ of a triangle ABC are divided by the points D,E,F in the same ratio, then the centroids of $\Delta ABC, \Delta DEF$ are coincide.

Proof : Let the points D,E,F divides $\overline{BC}$, $\overline{CA}$ and $\overline{AB}$ in the ratio of 1 : λ respectively.

$$D=\left(\frac{\lambda x_2+x_3}{\lambda+1},\frac{\lambda y_2+y_3}{\lambda+1}\right),$$

$$E=\left(\frac{\lambda x_3+x_1}{\lambda+1},\frac{\lambda y_3+y_1}{\lambda+1}\right)$$

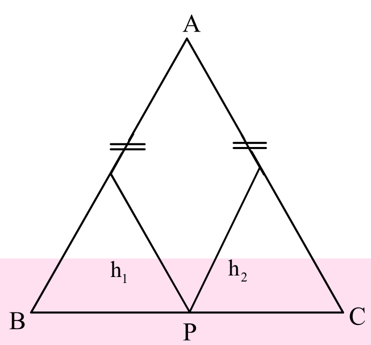
$$F=\left(\frac{\lambda x_1+x_2}{\lambda+1},\frac{\lambda y_1+y_2}{\lambda+1}\right)$$

Centroid of $\Delta\, DEF$ = Centroid of $\Delta\, ABC$

$$=\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$$

➢ In an isosceles triangle the sum of the distances from any point of the base to the lateral sides is constant.

Proof :



Let P be a point of $\overline{BC}$,

Let h_1, h_2 are $\perp^r$ distances from P to $\overline{AB}, \overline{AC}$ area of ΔABP, $\Delta_1=\frac{1}{2}AB\,h_1$

area of ΔACP, $\Delta_2=\frac{1}{2}AC\,h_2$

$$h_1+h_2=\frac{2}{AC}(\Delta_1+\Delta_2)=\frac{2\Delta}{AC}=\frac{2}{AC}\cdot\frac{1}{2}(AC)h$$

$h_1+h_2=h$ (h is altitude from B to $\overline{AC}$) sum of the distances is equal to the length of altitude drawn to a lateral side of the triangle.

➢ The line in the family of lines $L_1+\lambda L_2=0$ which is at maximum distance from a point P is perpendicular to $\overline{PA}$, where A is point concurrence of the family of lines.

C.U.Q

1. The straight line through A(a,b) intersects the line through B(c,d) at ‘P’ at right angles. The locus of P is

   1) (x–a)(x–c)+(y–b)(y–d)=0

   2) (x–a)(x–c)-(y–b)(y–d)=0

   3) (x–b)(x–d)+(y–a)(y–d)=0

   4) (x–b)(x–d)+(y–a)(y–c)=0

2. If ax+by+c=0 is parallel to x-axis then which of the following is defined

   1) $\frac{a^2-c^2}{c^2+b^2}$ 2) $\frac{b^2-c^2}{a^2}$ 3) $\frac{4b^2+c^2}{\sqrt{abc}}$ 4) $\frac{\sqrt{a+c^2}}{a}$

3. The straightline $ax+by+c=0\,(a,b,c\neq 0)$ will pass through the first quadrant and cut the positive x-axis, if

   1) $ac>0, bc>0$ 2) $ac>0, bc<0$

   3) $ac>0$ and / or $bc>0$

4) $ac < 0$ and / or $bc < 0$

4 If $\frac{ax}{\sqrt{a^2+b^2}}+\frac{by}{\sqrt{a^2+b^2}}=\frac{-c}{\sqrt{a^2+b^2}}$ is perpendicular form of a straight line then

1) $a,b,c>0$ 2) $a>0,b>0,c<0$

3) $c<0$ 4) $a,b>0,c\neq 0$

5. The straight line passing through $P(x_1,y_1)$ and making an angle α with x-axis intersects Ax+By+C=0 in Q then PQ=

1) $\frac{|Ax_1+By_1+C|}{\sqrt{A^2+B^2}}$ 2) $\frac{|Ax_1+By_1+C|}{\sqrt{A\cos\alpha+B\sin\alpha}}$

3) $\frac{|Ax_1+By_1+C|}{|A\cos\alpha+B\sin\alpha|}$ 4) $\frac{|Ax_1+By_1+C|}{\sqrt{A^2\cos^2\alpha+B^2\sin^2\alpha}}$

6. If $a+b+c\neq 0$, ax+by+c=0 bx+cy+a=0, cx+ay+b=0 are concurrent then $\frac{a^2+b^2+c^2}{ab+bc+ca}=$

1) 1/2 2) 2 3) 1 4) 0

7. The lines

(a+b-2c)x+(b+c-2a)y+(c+a-2b)=0,

(b+c-2a)x+(c+a-2b)y+(a+b-2c)=0 and

(c+a-2b)x+(a+b-2c)y+(b+c-2a)=0 where a,b,c,real numbers

1) Form an equilateral triangle

2) Concurrent

3) Form a right angled triangle

4) Form an isosceles triangle

8. If the lines

$p_1x+q_1y=1, p_2x+q_2y=1 and\ p_3x+q_3y=1$

be concurrent, then the points $(p_1,q_1),(p_2,q_2)$ and (p_3,q_3),

1) are collinear

2) form an equilateral triangle

3) form a scalene triangle

4) form a right angled triangle

9. $m\equiv a_1x+b_1y+c_1=0$ and $l\equiv a_2x+b_2y+c_2=0$ are two straight lines such that $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$ then m + kl = 0, $k\in R$ is

1) a straight line different from m and l

2) not a straight line

3) is a straight line concurrent with m and l

4) the same straight line m = 0

10. If a and b are the intercepts made by the straight line on the coordinate axes such that $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}$ then the line passes through point

1) (1,1) 2) (c,c) 3) $\left(\frac{1}{c},\frac{1}{c}\right)$ 4) $\left(\frac{c}{a},\frac{c}{a}\right)$

11. The orthocentre of the triangle formed by the points $A(a\cos\alpha,a\sin\alpha)$ $B(a\cos\beta,a\sin\beta)$ $C(a\cos\gamma,a\sin\gamma)$ is

1) $(\cos\alpha+\cos\beta+\cos\gamma, \sin\alpha+\sin\beta+\sin\gamma)$

2) $[a(\cos\alpha+\cos\beta+\cos\gamma),a(\sin\alpha+\sin\beta+\sin\gamma)]$

3. $[a(\cos\alpha+\sin\beta+\sin\gamma),a(\sin\alpha+\cos\beta+\cos\gamma)]$

4. $(\cos\alpha\ \cos\beta\ \cos\gamma, \sin\alpha\ \sin\beta\ \sin\gamma)$

12. (a, b), (c, d), (e, f) are the vertices of an equilateral triangle. Then the orthocentre of the triangle is

1) $\left(\frac{a+d+f}{3},\frac{b+c+e}{3}\right)$

2) $\left(\frac{a+c+e}{3},\frac{b+d+f}{3}\right)$

3) $\left(\frac{a+c+f}{3},\frac{b+d+e}{3}\right)$

4) $\left(\frac{a+b+c}{3},\frac{d+e+f}{3}\right)$

13. A triangle is formed by the lines ax+by+c=0 lx+my+n=0, px+qy+r=0, then the straight line (ax+by+c)(*l*p+mq)= (*l*x+my+n) (*a*p+bq) passes through .... of the triangle.

1) Incentre 2) Circumcentre
3) Orthocentre 4) Centroid

14. The area of the triangle formed by the lines $y = m_1x + c_1, y = m_2x+c_2$ and x=0 is

1) $\frac{|c_1-c_2|}{2|m_1-m_2|}$ 2) $\frac{(c_1-c_2)^2}{2(m_2-m_1)^2}$

3) $\frac{(c_1-c_2)^2}{2|m_1-m_2|}$ 4) $\frac{(c_1-c_2)^2}{|m_2-m_1|}$

C.U.Q - KEY

1) 1 2) 1 3) 4 4) 3 5) 3
6) 3 7) 2 8) 1 9) 4 10) 2
11) 2 12) 2 13) 3 14) 3

C.U.Q - HINTS

1. Locus of P is a circle with end points of the diameter A,B
2. Line parellel to x-axis
$\Rightarrow$x-coeffient =0
3. $-\frac{c}{a}>0$
4. $x\cos\alpha + y\sin\alpha = p, where \quad p>0$
5. let PQ = r
$Q\left(x_1 \pm r\cos\theta, y_1 \pm r\sin\theta\right)$ sub in Ax+By+C=0 .
6. $a^3+b^3+c^3-3abc=0$
$\left(a+b+c\right)\left(a^2+b^2+c^2-ab-bc-ca\right)=0$
7. $L_1+L_2+L_3=0$
8. Area = 0
9. $m=0, l=0$ represent coincident lines
10. $\frac{c}{a}+\frac{c}{b}=1$
11. If S= 0 then H = 3G
12. S = G
13. Given equation is altitude
14. Use area of triangle formula

LEVEL - I (C.W)

Slope of a line:

1. If the line passing through the points (-8,3) (2,1) is parallel to the line passing through the points (11,-1) (k,0) then the value of k is

1) 5 2) 7 3) 5/2 4) 6

2. If each of the points (a,4),(-2,b) lies on the line joining the points (2,-1),(5,-3),then the point (a,b) lies on line

1) 6x+6y-25=0 2) x+3y+1=0
3) 2x+6y+1=0 4) 2x+3y-5=0

3. If the lines y = – 3x + 4, ay = x + 10 and 2y + bx + 9 = 0 represent three consecutive sides of a rectangle then ab =

1) 18 2) –3 3) $\frac{1}{2}$ 4) $\frac{-1}{3}$

Slope-intercept form,slope-point form and two-point form:

4. The equation of the stratight line cutting off an intercept 8 on x–axis and making an angle of 60^0 with the positive direction of y –axis is

1) $x-\sqrt{3}y-8=0$ 2) $x+\sqrt{3}y=8$
3) $y-\sqrt{3}x=8$ 4) $y+\sqrt{3}x=8$

5. If (-4,5) is one vertex and 7x-y+8=0 is one diagonal of a square, then the equation of the other diagonal is

1) x+7y-31=0 2) x+7y-15=0
3) x+7y+8=0 4) x+7y-35=0

Intercepts and intercept form:

6. Equation of a line which passes through the point (-3,8) and cut off positive intercepts on the axes whose sum is 7 is

1) 3x-4y=12 2) 4x+3y=12
3) 3x+4y=12 4) 4x-3y=12

7. The number of lines that are parallel to 2x + 6y– 7 = 0 and have an intercept 10 between the co-ordinate axes is

1) 1 2) 2 3) 4 4) infinitely many

8. If the line (x-y+1) + k (y-2x+4) = 0 makes

equal intercept on the axes then the value of k is

1) $1/3$ 2) $3/4$ 3) $1/2$ 4) $2/3$

Normal form and symmetric form:

9. **The equation of set of lines which are at a constant distance 2 units from the origin is**

1) x+y+2=0
2) x+y+4=0
3) $x\cos\alpha + y\sin\alpha = 2$
4) $x\cos\alpha + y\sin\alpha = ½$

10. **The slope of a straight line through A(3,2) is 3/4 then the coordinates of the two points on the line that are 5 units away from A are**

1) (-7,5) (1,-1) 2) (7,5) (-1,-1)
3) (6,9) (-2,4) 4) (7,3) (-2,1)

Problems on distances:

11. **The length of the perpendicular from the point (0,0) to the straight line passing through P(1,2) such that P bisects the intercepted part between axes**

1) $\sqrt{5}$ 2) 4 3) $4/\sqrt{5}$ 4) $\sqrt{5}/4$

12. **Radius of the circle touching the lines 3x+4y-14=0, 6x+8y+7=0 is (EAM- 2011)**

1) 7 2) $\frac{7}{2}$ 3) $\frac{7}{4}$ 4) $\frac{7}{6}$

13. **The distance between the parallel lines given by $(x+7y)^2 + 4\sqrt{2}(x+7y) - 42 = 0$ is (EAM- 2012)**

1) 1 2) 5 3) 6 4) 2

14. **Equation of the straight line parallel to x+2y-5=0 and at the same distance from (3,2) is**

1) x+2y-8=0 2) x+2y+9=0
3) x+2y-9=0 4) x+2y-7=0

15. **If the straight line drawn through the point $P(\sqrt{3},2)$ making an angle $\frac{\pi}{6}$ with x-axis meets the line $\sqrt{3}$x-4y+8=0 at Q. Then PQ is**

1) 4 2) 5 3) 6 4) 9

Position of a point (s) w.r.t. line (s):

16. **If the line 3x+4y-8=0 is denoted by L, then the points (2,-5),(-5,2)**

1) lie on L
2) lie on same side of L
3) lie on opposite sides of L
4) equidistant from L

17. **The vertices of a triangle are O(0.0), B(-3,-1),C(-1,-3). The equation of the line parallel to BC and intersecting the sides OB and OC whose perpendicular distance from O is 1/2 is**

1) $x + y = 1/\sqrt{2}$ 2) $x + y = -1/\sqrt{2}$
3) $x + y = -1/2$ 4) $x + y = 1/2$

Point of intersection of lines and Concurrency of Lines :

18. **If the lines ax+by+c = 0, bx+cy+a = 0 and cx+ay+b=0 $a \neq b \neq c$ are concurrent then the point of concurrency is**

1) (0,0) 2) (1,1) 3) (2,2) 4) (-1,-1)

19. **If the lines 3x+2y-5=0, 2x-5y+3=0, 5x+by+c=0 are concurrent then b+c =**

1) 7 2) -5 3) 6 4) 9

20. **The lines (p-q) x + (q-r) y + (r-p) = 0**
(q-r) x + (r-p) y + (p-q) = 0
(r-p) x + (p-q) y + (q-r) = 0

1) Form an equilateral triangle
2) Form an Isosceles triangle
3) are Concurrent
4) Form a right angled triangle

21. **If 2 is a root of $ax^2+bx+c=0$ then point of concurrence of lines ax+2by+3c=0 is**

1) (12,3) 2) (4,2) 3) (1,2) 4) (2, 3)

22. **For all values of 'a' the set of straight lines (3a+1) x - (2a+3) y + 9-a=0 passes through the point**

1) (3, 4) 2) (4,2) 3) (3,3) 4) (1,2)

23. **Equation of the line passing through the point of intersection of the lines 2x+3y-1=0, 3x+4y-6=0 and perpendicular to 5x-2y-7=0 is (EAM- 2009)**

1) 2x+5y-19=0 2) 2x+5y+17=0
3) 2x+5y-16=0 4) 2x+5y-22=0

Angle between lines:

24. **If $2x+3y+4=0$ & $\lambda x+ky+2=0$ are**

identical lines then $3\lambda - 2k =$

1) 1 2) 0 3) -1 4) 2

25. The angle between the diagonals of a quadrilateral formed by the lines $\frac{x}{a}+\frac{y}{b}=1$, $\frac{x}{b}+\frac{y}{a}=1$, $\frac{x}{a}+\frac{y}{b}=2$, $\frac{x}{b}+\frac{y}{a}=2$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

Triangles and area of the triangle:

26. The triangle formed by the lines $\sqrt{3}x+y-2=0$, $\sqrt{3}x-y+1=0$, y=0 is

1) Equilateral 2) Right angled
3) Right angled isosceles 4) Isosceles

27. The area of the triangle formed by the lines x=0; y=0 and $x\sin 18^0 + y\cos 36^0 + 1 = 0$ is

1)1 2)2 3)3 4)4

28. If a straight line perpendicular to 3x-4y-6=0 forms a triangle with the coordinate axes whose area is 6sq. units, then the equation of the straight line (s) is (EAM- 2014)

1) x-2y=6 2) 4x+3y=12
3) 4x+3y+24=0 4) 3x-4y=12

29. The equation of base of an equilateral triangle is x+y=2 and the vertex is (2, -1). Then area of triangle is

1) $2\sqrt{3}$ 2) $\sqrt{3}/6$ 3) $1\sqrt{3}$ 4) $2\sqrt{3}$

Quadrilaterals and area of the quadrilaterals:

30. The quadrilateral formed by the lines 2x-5y+7=0, 5x+2y-1=0, 2x-5y+2=0, 5x+2y+3=0 is

1) Rectangle 2) Square
3) Parallelogram 4) Rhombus

31. The diagonals of a parallelogram PQRS are along the lines $x+3y=4$ and $6x-2y=7$. Then $PQRS$ must be :

1) rectangle 2) square
3) cyclic quadrilateral 4) rhombus

Foot and Image:

32. Foot of the perpendicular of origin on the line joining the points $(a\cos\theta, a\sin\theta)$, $(a\cos\phi, a\sin\phi)$ is

1) $(\cos\theta+\cos\phi, \sin\theta+\sin\phi)$
2) $(\cos\theta-\cos\phi, \sin\theta-\sin\phi)$
3) $\left(\frac{a(\cos\theta+\cos\phi)}{2}, \frac{a(\sin\theta+\sin\phi)}{2}\right)$
4) $(\cos\theta\cos\phi, \sin\theta\sin\phi)$

33. Suppose A, B are two points on 2x-y+3=0 and P(1,2) is such that PA=PB. Then the mid point of AB is

1) $\left(\frac{-1}{5},\frac{13}{5}\right)$ 2) $\left(\frac{-7}{5},\frac{9}{5}\right)$

3) $\left(\frac{7}{5},\frac{-9}{5}\right)$ 4) $\left(\frac{-7}{5},\frac{-9}{5}\right)$

34. A line passing through the points $(7,2),(-3,2)$ then the image of the line in x-axis is

1) $y=4$ 2) $y=9$ 3) $y=-1$ 4) $y=-2$

35. Image of the curve $x^2+y^2=1$ in the line $x+y=1$ is

1) $x^2+y^2+2x+2y+1=0$
2) $x^2+y^2-2x+2y+1=0$
3) $x^2+y^2+2x-2y+1=0$
4) $x^2+y^2-2x-2y+1=0$

36. Image of (1,2) w.r.t. (-2,-1) is

1) (0,5) 2) (-4,-3) 3) (-5,-4) 4) (-4,-5)

37. The image of the point (-2,-7) under the transformation (x,y) $\rightarrow$ (x-2y,-3x+y) is

1) (–12,1) 2) (12,–1) 3)(–12,–1) 4) (12,1)

Centroid, circumcentre, orthocentre and incentre:

38. The algebraic sum of the perpendicular distances from the vertices of a triangle to a variable line is 'O', then the line passes through the ------ of the triangle

1) Incentre 2) Centroid
3) Orthocentre 4) Circumcentre

39. A(1,-1) B(4,-1) C(4,3) are the vertices of a

triangle. Then the equation of the altitude through the vertex 'A' is

1) x = 4 2) y = 4 3) y + 1= 0 4) x =1

40. The equations of the sides of a triangle are x-3y=0, 4x+3y=5, 3x+y=0. The line 3x-4y=0 passes through

1) Incentre 2) Centroid
3) Orthocentre 4) Circumcentre

41. Equation of a diameter of the circum circle of the triangle formed by the lines 3x+4y-7=0, 3x-y+5=0 and 8x-6y+1=0 is

1) 3x-y-5=0 2) 3x+y+5=0
3) 3x-y+5=0 4) 3x+y-5=0

42. The incentre of the triangle formed by the lines $x\cos\alpha + y\sin\alpha = \pi$, $x\cos\beta + y\sin\beta = \pi$, $x\cos\gamma + y\sin\gamma = \pi$ is (α,β) then $\alpha+\beta =$

1) 0 2) 1 3) 2 4) 4

LEVEL-I (C.W)-KEY

1) 4	2) 3	3) 1	4) 2	5) 1
6) 2	7) 2	8)4	9) 3	10) 2
11) 3	12) 3	13) 4	14) 3	15) 3
16) 2	17) 2	18) 2	19) 2	20) 3
21) 1	22) 1	23) 2	24) 2	25) 4
26) 1	27) 2	28) 2	29) 2	30) 1
31) 4	32) 3	33) 1	34) 4	35) 4
36) 3	37) 2	38) 2	39) 3	40) 3
41) 3	42) 1			

LEVEL-I (C.W)-HINTS

1. Slopes of the parallel lines are equal
2. a=-11/2, b=5/3
3. let the given sides are AB, BC, CD
   $AB \parallel CD \Rightarrow b = 6$
   $AB \perp BC \Rightarrow a = 3$
4. $y = m(x-8), m = \tan 150$
5. Write the equation to a line perpendicular to $7x - y + 8 = 0$ and sub $(-4,5)$
6. Verification
7. Two lines parallel to any given line make intercept of same length k between the axes in opposite quadrants
8. m = -1
9. p = 2, $x\cos\alpha + y\sin\alpha = 2$
10. $(x_1, y_1) = (3,2)$
    $\tan\theta = \frac{3}{4}$ $\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$
    $(x,y) = (x_1 \pm r\cos\theta, y_1 \pm r\sin\theta)$
11. Eq. of the given line is 2x+y=4
    required distance= $\frac{4}{\sqrt{5}}$
12. Distance between two parallel tangents of a circle = Diameter
13. Find the distance between
    $x + 7y - 3\sqrt{2} = 0$ and $x + 7y + 7\sqrt{2} = 0$
14. Verification
15. $\left|\frac{ax_1 + by_1 + c}{a\cos\alpha + b\sin\alpha}\right|$
16. Use $m : n = -L_{11} : L_{22}$
17. line parallel to $\overline{BC}$ is x+y=k
    from given data $\frac{|k|}{\sqrt{2}} = \frac{1}{2} \Rightarrow k = -\frac{1}{\sqrt{2}}$
    ($\because$ O,B are opposite sides of the line)
18. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$ a+b+c=0
19. $\begin{vmatrix} 3 & 2 & -5 \\ 2 & -5 & 3 \\ 5 & b & c \end{vmatrix} = 0$
20. $L_1 + L_2 + L_3 = 0$
21. 4a+2b+c=0 & ax+2by+3c=0
22. Solve $3x - 2y - 1 = 0$, $x - 3y + 9 = 0$
23. intersecting point (14,-9) , $m = \frac{-2}{5}$
24. $\frac{2}{\lambda} = \frac{3}{k} = \frac{4}{2}$
25. The quadrilateral formed by the lines is a rhombus

26. $\sqrt{a_1^2+b_1^2}\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} : \sqrt{a_2^2+b_2^2}\begin{vmatrix} a_3 & b_3 \\ a_1 & b_1 \end{vmatrix} : \sqrt{a_3^2+b_3^2}\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$

27. $\Delta = \dfrac{c^2}{2|ab|}$

28. The line perpendicular to given line is

$4x+3y+k=0$ $\quad \therefore \dfrac{k^2}{24}=6$

29. Area $= \dfrac{p^2}{\sqrt{3}}$ *where p is height of* Δ

30. Adjacent sides are perpendicular and distance between parallel sides are not equal.

31. Given equation of the diagonals of a parallelogram are at right angle. Hence it is a rhombus.

32. Mid point becuase $OA = OB$

33. Apply foot of the perpendicular formula

34. Line equation y =2 Image with respect to x-axex is y =-2

35. Image of (0,0) in line is (1,1)

$\therefore$ image circle is $(x-1)^2+(y-1)^2=1$

36. The image of (x_1,y_1) w.r.to(x,y) is $(2x-x_1, 2y-y_1)$

37. apply given condition

38. Algebric sum of the distances from the three non collinear points to variable line is zero then the line passing through centroid of the triangle formed by this points.

39. $AB \perp BC$

40. Given lines form a right angle triangle

41. Hypotenous is diameter

42. (0,0) is equidistance from sides

LEVEL - I (H.W)

Slope of a line:

1. If the slope of a line is $\dfrac{-1}{\sqrt{3}}$ then its inclination is

1) $\dfrac{\pi}{3}$ 2. $\dfrac{5\pi}{6}$ 3. $\dfrac{2\pi}{3}$ 4. $\dfrac{3\pi}{4}$

2. If the straight line (3x+4y+5)+k(x+2y-3)=0 is parallel to x-axis then the value of k is

1) 1 2) -3 3) 4 4) 2

3. Number of straight lines passing through (1, 3),(7, -3),(5, -1),(6, -2) is

1) 2 2) $4c_2$ 3) $4p_2$ 4) $4c_4$.

Slope-intercept form,slope-point form and two-point form:

4. The equation of the horizontal line passing through the point (4,-7) is

1) y-7=0 2) y+7=0 3) y-4=0 4) y+4=0

5. The equation of the straight line making an intercept of 3 units on the y-axis and inclined at 45^0 to the x-axis is

1) y = x-1 2) y= x+3
3) y = 45x + 3 4) y = x+45

Intercepts and intercept form:

6. Equation of the line having intercepts a,b on the axes such that a+b=5 and ab = $\dfrac{21}{4}$ is

1) 3x+2y=6 2) 2x+3y=6
3) 14x+6y=21 4) x+4y=4

7. x intercept of the line parallel to 4x+7y=9 and passing through (2,3) is

1) $\frac{25}{4}$ 2) $\frac{17}{4}$ 3) $\frac{29}{4}$ 4) $\frac{29}{7}$

8. A straight line meet the axes in A and B such that the centroid of triangle OAB is (a,a). Then the equation of the line AB is

1) x+y=a 2) x-y=3a
3) x+y=2a 4) x+y=3a

Normal form and symmetric form:

9. Equation of the line on which the length of the perpendicular from origin is 5 and the angle which this perpendicular makes with the x axis is 60^O

1) $x+\sqrt{3}y=12$ 2) $\sqrt{3}\,x+y=10$
3) $x+\sqrt{3}y=8$ 4) $x+\sqrt{3}\,y=10$

10. **A point on line $x-y+1=0$ at a distance $2\sqrt{2}$ from the point $(1,2)$ is**
1) $(3,4)$ 2) $(3,0)$ 3) $(-1,4)$ 4) $(0,1)$

Problems on distances:

11. **The perpendicular distance from (1,2) to the straight line 12x+5y=7 is**
1) 15/13 2) 12/13 3) 5/13 4) 7/13

12. **The vertices of a triangle are A(5,6) B(1, -4) C(-4,0) then the length of the altitude through the vertex A is**
1) $\frac{66}{\sqrt{41}}$ 2) $\frac{55}{\sqrt{41}}$ 3) $\frac{17}{5}$ 4) $\frac{19}{5}$

13. **The distance between the parallel lines 8x+6y+5=0 and 4x+3y-25=0 is**
1) $\frac{7}{2}$ 2) $\frac{9}{2}$ 3) $\frac{11}{2}$ 4) $\frac{5}{4}$

14. **The equation of the line which is parallel to 5x+12y+1=0 and 5x+12y+7=0 and lying midway between them is**
1) 5x+12y+13=0 2) 5x+12y-4=0
3) 5x+12y+4=0 4) 5x+12y-6=0

15. **The point on the line x + y = 4 that lies at a unit distance from the line 4x + 3y - 10 = 0 is**
1) (1,3) 2) (-7,11) 3) (11,-7) 4.(2,2)

Position of a point (s) w.r.t. line (s):

16. **The ratio in which the line 3x+4y-7=0 divides the line joining the points (1,2) (2,3) is**
1) 4:11 Internally 2) 4:11 Externally
3) 7:11 Internally 4) 7:11 Externally

17. **The line segment joining the points (1,2) and (k,1) is divides by the line $3x+4y-7=0$ in the ratio 4:9 then k is**
1) 2 2) -2 3) 3 4) -3

Point of intersection of lines and Concurrency of Lines:

18. **If the point of intersection of kx+4y+2=0, x-3y+5=0 lies on 2x+7y-3=0 then k=**
1) 2 2) 3 3)-2 4) -3

19. **The lines $px+qy+r=0$, $qx+ry+p=0$, $rx+py+q=0$, are concurrant then**
1) $p+q+r=0$ 2) $p^3+q^3+r^3=3pqr$
3) $p^2+q^2+r^2-pq-qr-rp=0$
4) All the above

20. **The point of concurrence of the lines $\frac{x}{3}+\frac{y}{4}=1$, $\frac{x}{4}+\frac{y}{3}=1$, x = y is**
1) $\left(\frac{4}{3},\frac{4}{3}\right)$ 2) $\left(\frac{2}{7},\frac{2}{7}\right)$
3) $\left(\frac{12}{7},\frac{12}{7}\right)$ 4) $\left(\frac{7}{12},\frac{7}{12}\right)$

21. **If 4a+5b+6c=0 then the set of lines ax+by+c=0 are concurrent at the point**
1) $\left(\frac{2}{3},\frac{5}{6}\right)$ 2) $\left(\frac{1}{3},\frac{1}{2}\right)$ 3) $\left(\frac{1}{2},\frac{4}{3}\right)$ 4) $\left(\frac{1}{3},\frac{7}{3}\right)$

22. **Let a and b be nonzero reals . Then the equation of the line passing through the origin and the point of inter section of x/a + y/b =1 and x/b + y/a=1**
1) ax+by=0 2) bx+ay=0
3) y-x=0 4) x+y=0

23. **The line which is concurrent with the lines 2x + 3y = 7, 2x = 3y + 1 and passing through the origin is**
1) $x+2y=0$ 2) $x-2y=0$
3) $2x+y=0$ 4) $2x-y=0$

Angle between lines:

24. **If θ is an acute angle between the lines y=2x+3, y=x+1 then the value of tan θ =**
1) $\frac{2}{3}$ 2) $\frac{1}{3}$ 3) $\frac{3}{4}$ 4) $\frac{1}{2}$

25. **The angle between the lines kx+y+9=0, y-3x=4 is 45^{O} then the value of k is(EAM- 2007)**
1) 2 or ½ 2) 2 or -1/2
3) -2 of ½ 4) -2 or -1/2

Triangles and area of the triangle:

26. **The straight lines x + y = 0, 3x + y – 4 = 0 and x+3y–4=0 form a triangle which is**
1) isosceles 2) right angled
3) equilateral 4) right angled isosceles triangle

27. **If a, c, b are three terms of a G.P., then the line ax + by +c =0**
1) has a fixed direction
2) always passes through a fixed point
3) forms a triangle with the axes whose area is constant
4) always cuts intercepts on the axes such that their sum is zero

28. **Area enclosed by the co-ordinate axes and the line passing through the pionts (8,-3), (-4,12) is**
1) $\frac{98}{5}$ 2) $\frac{49}{5}$ 3) $\frac{24}{25}$ 4) $\frac{17}{8}$

29. **A straight line L is perpendicular to the line 4x-2y=1 and forms a triangle of area 4 sq.units with coordinate axes then, an equation of L is**
1) 2x+4y+8=0 2) 2x-4y+8=0
3) 2x+4y+7=0 4) 4x-2y-7=0

Quadrilaterals and area of the quadrilaterals:

30. **A square of area 25 sq.units is formed by taking two sides as $3x+4y=k_1$ and $3x+4y=k_2$ then $|k_1-k_2|=$**
1) 5 2) 1 3) 25 4) 125

31. **The equations of two sides of a square whose area is 25 sq.units are 3x-4y=0 and 4x+3y=0. The equation of other two sides of the square are**
1) $3x-4y\pm 25=0, 4x+3y\pm 25=0$
2) $3x-4y\pm 5=0, 4x+3y\pm 5=0$
3) $3x-4y\pm 5=0, 4x+3y\pm 25=0$
4) $3x-4y=0, 4x+3y=0$

Foot and Image:

32. **If 2x+3y=5 is the perpendicular bisector of the line segment joining the points A $(1,\frac{1}{3})$ and B then B=** **(EAM- 2013)**
1) $\left(\frac{21}{13},\frac{49}{39}\right)$ 2) $\left(\frac{17}{13},\frac{31}{39}\right)$
3) $\left(\frac{7}{13},\frac{49}{39}\right)$ 4) $\left(\frac{21}{13},\frac{31}{39}\right)$

33. **If (2, -3) is the foot of the perpendicular from (-4, 5) on a line then its equation is**
1) 3x-4y+28=0 2) 3x-4y-18=0
3) 3x-4y+18=0 4) 3x-4y-17=0

34. **If (-2, 6) is the image of the point (4,2) with respect to the line L=0, then L=** **(EAM- 2014)**
1) 6x-4y-7=0 2) 2x+3y-5=0
3) 3x - 2y+5=0 4) 3x-2y+10=0

35. **One vertex of a square ABCD is A(-1,1) and the equation of one diagonal BD is 3x+y-8=0 then C=**
1) (-5,3) 2) (5,3) 3) (-5,-3) 4) (2,5)

36. **The reflection of the point (6,8) in the line x = y is**
1) (4,2) 2) (-6,-8) 3) (-8,-10) 4) (8,6)

Centroid, circumcentre, orthocentre and incentre:

37. **The vertices of a triangle are (2,0) (0,2) (4,6) then the equation of the median through the vertex (2,0) is**
1) x+y-2=0 2)x=2
3) x+2y-2=0 4) 2x+y-4=0

38. **If the algebraic sum of the perpendicular distances from the points (2,0) (0,2) and (4,4) to a variable line is 'O', then the line passes through the fixed point**
1) (1,1) 2) (3,3) 3) (2,2) 4) (0,0)

39. **If the equations of the three sides of a triangle are $x+y=1, 3x+5y=2$ and $x-y=0$ then the orthocentre of the triangle lies on the line**
1) $5x-3y=2$ 2) $3x-5y-1=0$
3) $2x-3y=1$ 4) $5x-3y=1$

40. **Circum centre of the triangle formed by the lines x+y=0, 2x+y+5=0, x-y=2 is**

1) (-2, -1) 2) (-3, 1)
3) (-4, 3) 4)(-1, -3)

41. The incentre of the triangle formed by the lines 3x + 4y =10, 5x + 12y =26, 7x+24y=50 is (α,β) then $\alpha+\beta$ =

1) 0 2) 1 3) 2 4) 4

LEVEL-I(H.W) - KEY

1) 2	2) 2	3) 4	4) 2	5) 2
6) 3	7) 3	8) 4	9) 4	10) 1
11) 1	12) 1	13) 3	14) 3	15) 2
16) 2	17) 2	18) 2	19) 4	20) 3
21) 1	22) 3	23) 2	24) 2	25) 2
26) 3	27) 3	28) 1	29) 1	30) 3
31) 1	32) 1	33) 2	34) 3	35) 2
36) 4	37) 2	38) 3	39) 4	40) 2
41) 1				

LEVEL-I(H.W) - HINTS

1. $\tan\theta = \frac{-1}{\sqrt{3}}$

2. Coefficient of x = 0

3. Given points are collinear

4. $y = y_1$

5. $y = mx + c$

6. a=3/2, b=7/2

7. $y-3=\frac{-4}{7}(x-2)$, Put y = 0

8. Intercepts α,β

$\therefore\left(\frac{\alpha}{3},\frac{\beta}{3}\right)=(a,a)$

9. $P=5,\alpha=60^0$

$x\cos\alpha + y\sin\alpha = P$

10. $P(x_1 \pm r\cos\theta, y_1 \pm r\sin\theta)$ with

$r=2\sqrt{2},\theta=\pi/4$

11. The perpendicular distance from (x_1,y_1) to the st line $ax+by+c=0$ is $\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$

12. Find the perpendicular distance from A to BC

13. $d=\frac{|c_1-c_2|}{\sqrt{a^2+b^2}}$

14. $ax+by+\frac{c_1+c_2}{2}=0$

15. Verification

16. Use $m:n=-L_{11}:L_{22}$

17. $-\frac{L_{11}}{L_{22}}=\frac{4}{9}\Rightarrow 3-3k=9$

18. $\begin{vmatrix} k & 4 & 2 \\ 1 & -3 & 5 \\ 2 & 7 & -3 \end{vmatrix}=0$

19. $\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix}=0$

$(p+q+r)(p^2+q^2+r^2-pq-qr-rp)=0$

20. Point of intersection is $\left(\frac{12}{7},\frac{12}{7}\right)$

21. $a\left(\frac{4}{6}\right)+b\left(\frac{5}{6}\right)+c=0$

22. Intersecting point of $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$ is $\left(\frac{ab}{a+b},\frac{ab}{a+b}\right)$

23. point of intersection of given lines is (2,1)

24. $\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$

25. $\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=1$

26. $\sqrt{1+1}\begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}:\sqrt{3^2+1^2}\begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix}:\sqrt{1^2+3^2}\begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix}$

27. $c^2=ab$

$\Delta=\frac{c^2}{2ab}=\frac{1}{2}$.

28. The line equation is

$5x+4y-28=0$, $\Delta=\dfrac{c^2}{2|ab|}$

29. 2x+y+k=0 forms a triangle of area 4.

30. $\left|\dfrac{k_1-k_2}{5}\right|=5 \Rightarrow |k_1-k_2|=25$

31. Verification

32. B is image of A

33. Line is perpendicular to AB

34. Required line is the perpendicular bisector of given points.

35. Image of A w.r.to diagonal BD is C

36. The reflection of the point (a,b) in the line x=y is (b,a)

37. $A(2,0)B(0,2)C(4,6)$
mid point of BC is $D(2,4)$
Equation of AD is $x=2$

38. Centroid

39. given lines form a right angled triangle

40. Given lines form a right angle triangle

41. Incentre is equidistance from sides,
Hence I = (0 ,0)

LEVEL-II (C.W)

Slope of a line:

1. The lines $p(p^2+1)x-y+q=0$ and $(p^2+1)^2x+(p^2+1)y+2q=0$ are perpendicular to a common line for [AIEEE - 2009]

1) exactly one value of p
2) exactly two values of p
3) more than two values of p
4) no values of p

2. The slope of the line passing through the points $(2,\sin\theta),(1,\cos\theta)$ is 0 then general solution of θ

1) $n\pi+\dfrac{\pi}{4},\forall n\in Z$ 2) $n\pi-\dfrac{\pi}{4},\forall n\in Z$

3) $n\pi\pm\dfrac{\pi}{4},\forall n\in Z$ 4) $n\pi,\forall n\in Z$

Slope-intercept form,slope-point form and two-point form:

3. The perpendicular bisector of the line segment joining $P(1,4)$ and $Q(K,3)$ has Y intercept -4. then a possible value of K is (AIEEE-2008)

1) -4 2) 1 3) 2 4) -2

4. $P(\alpha,\beta)$ lies on the line y=6x-1 and Q(β,α) lies on the line 2x-5y=5. Then the equation of the line $\overleftrightarrow{PQ}$ is

1) 2x+y=3 2) 3x+2y=5 3) x+y=6 4)3x+y=7

5. A line joining A(2,0) and B(3,1) is rotated about A in anticlock wise direction through angle 15°, then the equation of AB in the new position is

1) $y=\sqrt{3}\,x-2$ 2) $y=\sqrt{3}\,(x-2)$
3) $y=\sqrt{3}\,(x+2)$ 4) $x-2=\sqrt{3}\,y$

Intercepts and intercept form:

6. The line 2x+3y=6, 2x+3y=8 cut the X-axis at A,B respectively. A line L = 0 drawn through the point (2,2) meets the X-axis at C in such a way that abscissa of A,B,C are in arithmetic Progression. then the equation of the line L is

1) 2x+3y=10 2) 3x+2y=10
3) 2x-3y=10 4) 3x-2y=10

7. The sum of the intercepts cut off by the axes on lines
$x+y=a$, $x+y=ar$, $x+y=ar^2$,............
where $a\neq 0$ and $r=\dfrac{1}{2}$

1) $2a$ 2) $a\sqrt{2}$ 3) $2\sqrt{2}a$ 4) a

8. The equation of the straight line which bisects the intercepts between the axes of the lines *x + y = 2 and 2x + 3y = 6 is*

1) $2x=3$ 2) $y=1$ 3) $2y=3$ 4) $x=1$

9. Equation of the line passing through (0, 1) and having intercepts in the ratio 2 : 3 is

1) 2x + 3y = 3 2) 2x – 3y + 3 = 0
3) 3x + 2y = 2 4) 2x – 3y – 3 = 0

Normal form and symmetric form:

10. A straight line is such that its distance of 5

units from the origin and its inclination is 135^{O}. The intercepts of the line on the co-ordinate axes are

1) 5, 5 2) $\sqrt{2},\sqrt{2}$

3) $5\sqrt{2},5\sqrt{2}$ 4) $5/\sqrt{2},5/\sqrt{2}$

11. Angles made with the x - axis by two lines drawn through the point (1, 2) and cutting the line x + y = 4 at a distance $\sqrt{\frac{2}{3}}$ from the point (1,2) are

1) $\frac{\pi}{6}$ and $\frac{\pi}{3}$ 2) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$

3) $\frac{\pi}{12}$ and $\frac{5\pi}{12}$ 4) $\frac{\pi}{4}$ *and* $\frac{\pi}{2}$

Problems on distances:

12. Perpendicular distance from the origin to the line joining the points $(a\cos\theta, a\sin\theta)$ $(a\cos\phi, a\sin\phi)$ is

1) $2a\cos(\theta-\phi)$ 2) $a\cos\left(\frac{\theta-\phi}{2}\right)$

3) $4a\cos\left(\frac{\theta-\phi}{2}\right)$ 4) $a\cos\left(\frac{\theta+\phi}{2}\right)$

13. One side of an equilateral triangle is 3x+4y=7 and its vertex is (1,2). Then the length of the side of the triangle is

1) $\frac{4\sqrt{3}}{17}$ 2) $\frac{3\sqrt{3}}{16}$ 3) $\frac{8\sqrt{3}}{15}$ 4) $\frac{4\sqrt{3}}{15}$

14. Equation of the line through the point of intersection of the lines 3x+2y+4=0 and 2x+5y-1=0 whose distance from (2,-1) is 2.

1) 2x-y+5=0 2) 4x+3y+5=0

3) x+2=0 4) 3x+y+5=0

15. If p,q denote the lengths of the perpendiculars from the origin on the lines $x\sec\alpha - y\operatorname{cosec}\alpha = a$ and $x\cos\alpha + y\sin\alpha = a\cos 2\alpha$ then (Eam 2013)

1) $4p^2+q^2=a^2$ 2) $p^2+q^2=a^2$

3) $p^2+2q^2=a^2$ 4) $4p^2+q^2=2a^2$

16. The distance between two parallel lines is p^1-p equation of one line is $x\cos\alpha + y\sin\alpha = p$ then the equation of the 2^{nd} line is

1) $x\cos\alpha + y\sin\alpha + p^1 + 2p = 0$

2) $x\cos\alpha + y\sin\alpha = 2p^1 - p$

3) $x\cos\alpha + y\sin\alpha = 0$

4) $x\cos\alpha + y\sin\alpha + p^1 - 2p = 0$

17. The ratio in which the line 3x+4y+2=0 divides the distance between 3x+4y+5=0 and 3x+4y-5=0 is

1) 7 : 3 2) 3 : 7 3) 2 : 3 4) 3 : 4

18. The equations of the lines parallel to 4x + 3y + 2 = 0 and at a distance of '4' units from it are

1. $4x + 3y + 22 = 0,\ 4x + 3y - 20 = 0$
2. $4x + 3y + 22 = 0,\ 4x + 3y - 18 = 0$
3. $4x + 3y - 18 = 0,\ 4x + 3y - 20 = 0$
4. $4x - 3y - 18 = 0,\ 4x + 3y - 20 = 0$

Position of a point (s) w.r.t. line (s):

19. The range of α for which the points $(\alpha,\alpha+2)$ and $\left(\frac{3\alpha}{2},\alpha^2\right)$ lie on opposite sides of the line $2x+3y-6=0$

1) $(-\infty,-2)$ 2) $(0,1)$

3) $(-\infty,-2)\cup(0,1)$ 4) $(-\infty,1)\cup(2,\infty)$

20. If $P\left(1+\frac{t}{\sqrt{2}},2+\frac{t}{\sqrt{2}}\right)$ be any point on a line then the range of values of t for which the point P lies between the parallel lines $x+2y=1$ *and* $2x+4y=15$ is

1) $-\frac{4\sqrt{2}}{5}<t<\frac{5\sqrt{2}}{6}$ 2) $-\frac{4\sqrt{2}}{3}<t<\frac{5\sqrt{2}}{6}$

3) $t<\frac{-4\sqrt{2}}{3}$ 4) $t<\frac{5\sqrt{2}}{6}$

21. A point which lies between 2x+3y–7=0 and 2x+3y+12=0 is

1) (5, 1) 2) (-1, 3) 3) (3, -5) 4) (7, -1)

22. A line L cuts the sides AB, BC of ΔABC in the ratio 2 : 5, 7 : 4 respectively. Then the line L cuts CA in the ratio

1) 7 : 10 2) 7 : –10 3) 10 : 7 4) 10 : –7

Point of intersection of lines and

concurrency of Lines :

23. **The number of integral values of m for which x-coordinate of point of intersection of the lines 3x+4y=9 and y = mx +1 is also an integer is**

1) 2 2) 0 3) 4 4) 11

24. **The line parallel to the x-axis and passing through the intersection of the lines ax+2by+3b=0 and bx – 2ay – 3a = 0, where (a, b) $\neq$ (0, 0) is**

(1)Above the x-axis at a distance of 3/2 from it
(2)Above the x-axis at a distance of 2/3 from it
(3)Below the x-axis at a distance of 3/2 from it
(4)Below the x-axis at a distance of 2/3 from it

25. **If a,b,c form a G P with common ratio r, the sum of the ordinates of the points of intersection of the line ax+by+c=0 and the curve $x+2y^2 = 0$ is**

1) $\frac{-r}{2}$ 2) $\frac{-r^2}{2}$ 3) $\frac{r}{2}$ 4) $\frac{r^2}{2}$

26. **Consider a family of straight lines $(x+y)+\lambda(2x-y+1)=0$. Find the equa tion of the straight line belonging to this family that is farthest from $(1,-3)$.**

1) $3x-3y+2=0$ 2) $6x+15y-7=0$
3) $5x+2y+1=0$ 4) $6x-15y+7=0$

27. **If the line $x=a+m$, $y=-2$ and $y=mx$ are concurrent , then least value of $|a|$ is**

1) 0 2) $\sqrt{2}$ 3) $2\sqrt{2}$ 4) 2

28. **If $a \neq b \neq c$, if $ax+by+c=0$ $bx+cy+a=0$ and $cx+ay+b=0$ are concurrent. Then the value of**

$2^{a^2b^{-1}c^{-1}}\ 2^{b^2c^{-1}a^{-1}}2^{c^2a^{-1}b^{-1}}$

1) 1 2) 4 3) 8 4) 16

29. **Line ax+ by + p =0 makes angle $\pi/4$ with $x\cos\alpha+y\sin\alpha=p, p\in R^+$. If these lines and the line $x\sin\alpha-y\cos\alpha=0$ are concurrent, then**

1) $a^2+b^2=1$ 2) $a^2+b^2=2$
3) $2(a^2+b^2)=1$ 4) $a^2-b^2=1$

Angle between lines:

30. **If p, q, r are distinct, then (q-r)x + (r-p) y + (p-q)=0 and (q^3-r^3) x+ (r^3-p^3) y + (p^3-q^3) = 0 represents the same line if**

1) p+q+r=0 2) p=q=r
3) $p^2+q^2+r^2=0$ 4) $p^3+q^3+r^3=0$

31. **The lines $(a+b)x+(a-b)y-2ab=0$, $(a-b)x+(a+b)y-2ab=0$ and $x+y=0$ form an isosceles triangle whose vertical angle is**

1) $\frac{\pi}{2}$ 2) $\tan^{-1}\left(\frac{2ab}{a^2-b^2}\right)$

3) $\tan^{-1}\left(\frac{a}{b}\right)$ 4) $2\tan^{-1}\left(\frac{a}{b}\right)$

32. **If 2(sina + sinb) x – 2sin (a – b) y = 3 and 2(cosa+cos b) x+2cos(a-b)y=5 are perpendicular then sin2a + sin 2b =**

1) sin (a-b) – 2sin (a+b)
2) sin 2(a-b) – 2sin (a+b)
3) 2sin (a-b) – sin (a+b)
4) sin2 (a-b) – sin (a+b).

33. **Two equal sides of an isoceles triangle are given by $7x-y+3=0$ and $x+y-3=0$ and the third side passes through the point (1, 10) then the slope m of the third side is given by**

1) $3m^2-1=0$ 2) $m^2+1=0$
3) $3m^2+8m-3=0$ 4) $m^2-3=0$

34. **The diagonal of a square is 8x – 15y = 0 and one vertex of the square is (1, 2). Then the equations to the sides of the square passing through the vertex are**

1. 22x + 8y = 9, 22x – 8y = 52

2. $23x + 7y = 9,\ 7x - 23y = 52$

3. $23x - 7y = 9,\ 7x + 23y = 53$

4. $22x - 8y = 9,\ 22x + 8y = 52$

Triangles and area of the triangle:

35. **Area of triangle formed by angle bisectors of coordinate axes and the line x=6 in sq.units is**

1) 36 2) 18 3) 72 4) 9

36. **The quadratic equation whose roots are the x and y intercepts of the line passing through (1,1) and making a traingle of area A with the co -ordinate axes is**

1) $x^2 + Ax + 2A = 0$ 2) $x^2 - 2Ax + 2A = 0$
3) $x^2 - Ax + 2A = 0$ 4) $(x - A)(x+A) = 0$

37. **A line passing through (3,4) meets the axes $\overline{OX}$ and $\overline{OY}$ at A and B respectively. The minimum area of the triangle OAB in square units is**

1) 8 2) 16 3) 24 4) 32

Quadrilaterals and area of the quadrilaterals:

38. **The figure formed by the straight lines $\sqrt{3}x + y = 0$, $\sqrt{3}y + x = 0$, $\sqrt{3}x + y = 1$, $\sqrt{3}y + x = 1$ is**

1) a rectangle 2) a square
3) a rhombus 4) parallelogram

39. **Let the base of a triangle lie along the line x = a and be of length a. The area of this triangle is a², if the vertex lies on the line**

1) $x + a = 0$ 2) $x = 0$
3) $2x - a = 0$ 4) $x - a = 0$

40. **The area bounded by $y = |x| - 1$, $y = -|x| + 1$**

1) 1 2) 2 3) $2\sqrt{2}$ 4) 4

41. **The area enclosed by $2|x| + 3|y| \le 6$ is**

1) 3 sq. units 2) 4 sq. units
3) 12 sq. units 4) 24 sq. units

Foot and Image:

42. **The point on the line $3x - 2y = 1$ which is closest to the origin is**

1) $\left(\frac{3}{13}, \frac{2}{13}\right)$ 2) $\left(\frac{5}{11}, \frac{2}{11}\right)$
3) $\left(\frac{3}{5}, \frac{2}{5}\right)$ 4) $\left(\frac{3}{13}, \frac{-2}{13}\right)$

43. **The reflection of y=$\sqrt{x}$ w.r.t. y-axis is**

1) $y = -\sqrt{x}$ 2) $y = \sqrt{-x}$
3) $y = -\sqrt{-x}$ 4) $x = \sqrt{y}$

44. **The points (-1, 1) and (1, -1) are symmetri cal about the line**

1) $y + x = 0$ 2) $y = x$
3) $x + y = 1$ 4) $x - y = 1$

45. **The equation of perpendicular bisectors of sides AB,BC of $\triangle$ABC are x-y-5=0, x+2y=0 respectively and A(1,-2) then coordinate of C are**

1) (1,0) 2) (0,1) 3) (5,0) 4) (0,0)

Centroid, circumcentre, orthocentre and incentre:

46. **If one vertex of an equilateral triangle is the origin and side opposite to it has the equation x+y=1, then the orthocentre of the triangle is**

1) $\left(\frac{1}{3}, \frac{1}{3}\right)$ 2) $\left(\frac{2}{3}, \frac{2}{3}\right)$ 3) (1,1) 4) (1,3)

47. **If the circum centre of the triangle lies at (0,0) and centroid is middle point of $\left(a^2+1, a^2+1\right)$ and (2a,-2a) then the orthocentre lies on**

1) $(a-1)^2 x - (a+1)^2 y = 0$

2) $(a-1)^2 x + (a+1)^2 y = 0$

3) $(a-1)^2 x + (a+1)^2 y + 56 = 0$

4) $(a-1)^2 x + (a+1)^2 y - 56 = 0$

48. **The orthocentre of the triangle formed by the lines x+y=1, 2x+3y=6 and 4x-y+9=0 lies in quadrant number**

1) 1st 2) IInd 3) IIIrd 4) IVth

49. **If the straight lines 2x+3y-1=0, x +2y-1=0 and ax + by –1 = 0 form a triangle with origin as orthocentre, then (a,b) is giveny by**

1) (6,4) 2) (-3,3) 3) (-8,8) 4) (0,7)

50. **In Δ ABC, equation to AB is 2x+3y-5=0, altitude through A is x-y+4=0 and altitude through B is 2x-y-1=0. Then the vertex C is**

1) $\left(-\frac{1}{5},\frac{9}{5}\right)$ 2) $\left(\frac{1}{5},\frac{9}{5}\right)$ 3) $\left(\frac{1}{5},\frac{-9}{5}\right)$ 4) $\left(\frac{-1}{5},\frac{-9}{5}\right)$

51. **Centroid of the triangle, formed by the lines x+2y-5=0, 2x+y-7=0, x-y+1=0 is**

1) (1,3) 2) (3,5) 3) (2,2) 4) (1,1)

Angular bisectors :

52. **The acute angle bisector between the lines 3x-4y-5=0, 5x+12y-26=0 is**

1) 7x-56y+32=0 2) 9x-3y+13=0
3) 14x-112y+65=0 4) 7x-13y+9=0

53. **The equation of the bisector of the angle between the lines x-7y+5=0, 5x+5y-3=0 which is the supplement of the angle containing the origin will be**

1) x+3y-2=0 2) x-3y+2=0
3) 3x-y+1=0 4) 3x+y+2=0

54. **Reflection of 3x+4y+5 =0 w.r.to the line 2x+y+1=0 is**

1. 2x+1=0 2. 2x-1=0
3. 5x-1=0 4. 5x+1=0

55. **Two sides of a Rhombus ABCD are parallel to the lines x-y=5 and 7x-y=3. The diagonals intersect at (2,1) then the equations of the diagonals are**

1) x-y=1, 7x-y=13 2) x+y=3,x+7y=9
3) x+2y=4, 2x-y=3 4) 3x+4y=10, 4x-3y=5

56. **Let P = (-1,0) Q=(0,0) and R=(3, 3$\sqrt{3}$) be three points. Then the equation of the bisector of angle PQR is (AIEEE 2007)**

1) $\frac{\sqrt{3}}{2}x + y = 0$ 2) $x + \sqrt{3}\,y = 0$

3) $\sqrt{3}\,x + y = 0$ 4) $x + \frac{\sqrt{3}}{2}y = 0$

Optimization and reflection in surface:

57. **A ray of light along $x+\sqrt{3}y=\sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is [JEE-MAINS 2013]**

1) $y = x+\sqrt{3}$ 2) $\sqrt{3}y = x-\sqrt{3}$
3) $y = 3x-\sqrt{3}$ 4) $\sqrt{3}y = x-1$

58. **Consider the points A(0,1) and B(2,0) and P be a point on the line $4x+3y+9=0$. Coordinates of P such that $|PA-PB|$ is maximum are**

1) $\left(\frac{-24}{5},\frac{17}{5}\right)$ 2) $\left(\frac{-84}{5},\frac{13}{5}\right)$

3) $\left(\frac{-6}{5},\frac{17}{5}\right)$ 4) $(0,-3)$

Miscellaneous problems:

59. **A straight line which make equal intercepts on +ve x and y axes and which is at a distance '1' unit from the origin intersects the straight line $y=2x+3+\sqrt{2}$ at (x_0,y_0) then $2x_0+y_0=$ [EAM 2010]**

1) $3+\sqrt{2}$ 2) $\sqrt{2}-1$ 3) 1 4) 0

60. **p is the length of the perpendicular drawn from the origin upon a straight line then the locus of mid point of the portion of the line intercepted between the coordinate axes is**

1) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{1}{p^2}$ 2) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{2}{p^2}$

3) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$ 4) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{1}{p}$

61. **Equation of the line passing through the point (2,3) and making intercept 2 units between the lines y+2x=3, y+2x=5 is**

1) x=2 2) y=3 3) x+y=5 4) x+y=7

62. **The number of lines that can be drawn through the point (4,-5) at a distance of 10 units from the point (1,3) is**

1) 0 2) 1 3) 2 4) Infinite

63. **The number of circles that touch all the 3 lines 2x + y = 3, 4x - y = 3, x + y = 2 is**

1) 0 2) 1 3) 2 4) 4

LEVEL-II (C.W)- KEY

1) 1	2) 1	3) 1	4) 3	5) 2
6) 1	7) 3	8) 2	9) 3	10) 3
11) 3	12) 2	13) 3	14) 2	15) 1
16) 4	17) 2	18) 2	19) 3	20) 2
21) 3	22) 4	23) 1	24) 3	25) 3
26) 4	27) 3	28) 3	29) 2	30) 1
31) 2	32) 2	33) 3	34) 3	35) 1

36) 2 37) 3 38) 3 39) 1 40) 2
41) 3 42) 4 43) 2 44) 2 45) 3
46) 1 47) 1 48) 2 49) 3 50) 2
51) 3 52) 3 53) 1 54) 3 55) 3
56) 3 57) 2 58) 1 59) 2 60) 3
61) 1 62) 1 63) 2

LEVEL-II (C.W)- HINTS

1. $\frac{a_1}{a_2}=\frac{b_1}{b_2}$
2. $\sin\theta=\cos\theta \quad \tan\theta=1$
3. $(0,-4)$ lies on perpendicular bisector $\overline{PQ}$
4. By solving $\beta=6\alpha-1$ and $2\beta-5\alpha=5$ we get P(1,5), Q(5,1)
5. New position of AB makes 15^0+45^0 inclination with x-axis
6. $A=(3,0)\,B(4,0)$; $c=\left(2-\frac{2}{m},0\right)$
7. Intercepts between the axes made by the given lines are $a\sqrt{2}, ar\sqrt{2}, ar^2\sqrt{2}$ ....
8. Midpoints of intercepts of given lines are (1,1), (3/2,1)
9. $mx+ny=mx_1+ny_1$
10. $\alpha=135^0-90^0$, P = 5
11. $r=\sqrt{2/3},(x_1,y_1)=(1,2)$
12. Find the distance between $(0,0)$ and midpoint of $(a\cos\theta,a\sin\theta)$ and $(a\cos\phi,a\sin\phi)$
13. $\frac{\sqrt{3}}{2}a=\frac{4}{5}$
14. Point of intersection $(-2,1)$ and verification
15. Put $\alpha=45$ and use perpendicular distance formula
16. Verify the distance between the parllel lines
17. distance between 3x+4y+2=0 & 3x+4y+5=0 distance between 3x+4y+2=0 & 3x+4y-5=0
18. $ax+by+c\pm d\sqrt{a^2+b^2}=0$
19. Points lie on opposite sides of the line $\Rightarrow L_{11}L_{22}<0$

$\Rightarrow 5\alpha(3\alpha+3\alpha^2-6)<0\to\alpha(\alpha+2)(\alpha-1)<0$

$\Rightarrow\alpha\in(-\infty,-2)\cup(0,1)$

20. Origin, P lies opposite side to the first line and same side to the second line
21. Verify $L_pL_p^1<0$ for options
22. $\left(\frac{BD}{DC}\right)\left(\frac{CE}{EA}\right)\left(\frac{AF}{FB}\right)=-1$
23. By solving, given equations we get $x=\frac{5}{3+4m}$

    x is an ineger of $3+4m=\pm1,\pm5$,

    $\therefore$ integral values of m are -1,-2
24. Eq. of required line parallel to x-axis

    $\Rightarrow$ slope $=0\Rightarrow\lambda=-a/b$

    Equation $=2y+3=0$
25. By solving two equations we get $2y^2-ry-r^2=0$
26. Line to perpendicular to line joining $(1,-3)$ and point of concurrency
27. by eliminating x,y from three equations we get – $2=m(a+m)\Rightarrow m^2+am+2=0$

    Since $m\in R\Rightarrow dis\geq0$

    $\therefore a^2-8\geq0\Rightarrow|a|\geq2\sqrt{2}$
28. $a+b+c=0$

    $a^3+b^3+c^3=3abc$
29. $ax+by+c=0$ is angle bisector of given two lines.
30. $\frac{q^3-r^3}{q-r}=\frac{r^3-p^3}{r-p}=\frac{p^3-q^3}{p-q}$
31. $\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$
32. $m_1m_2=-1$
33. $\frac{m-7}{1+7m}=-\frac{m+1}{1-m}\Rightarrow 3m^2+8m-3=0$
34. By using $\frac{m\pm\tan\alpha}{1\mp m\tan\alpha}$

    required slopes are 23/7,-7/23

35. Equations of the angular bisectors of the axes are $y = x$ and $y = -x$

36. $a+b=ab,\ \frac{1}{2}ab=A$

37. $(p,q)=(3,4)$ then minimum area = 2pq

38. Adj sides are not perpendicular and $d_1 = d_2$

39. Area $= a^2 \Rightarrow height\ of\ \Delta = 2a$

40. From the diagram required area $= 4(area\ of\ \Delta) = 4\left(\frac{1}{2}\right) = 2$

41. required area is $\frac{1}{2}\times 6\times 4 = 12$ sq. units

42. Foot of the perpendicular

43. The image of (x,y) w.r.to y-axis is $(-x,y)$

44. Required line is perpendicular bisector of $\overline{AB}$

45. B is image of A w.r.to x-y-5=0
C is image of B w.r.to x-2y=0

46. Foot of the perpendicular $D=\left(\frac{1}{2},\frac{1}{2}\right)$
G(=O) divides median in the ratio 2:1

47. Line joing circumcentre and centroid

48. apply $(a_1a_2+b_1b_2)\ L_3 = (a_1a_3+b_1b_3)$
$L_2 = (a_2a_3+b_2b_3)L_1$

49. Equation of AO is
$(2x+3y-1)+\lambda(x+2y-11)=0$ passes through (0,0) $\Rightarrow \lambda=-1$
Since $AO \perp BC$ we have a=-b
similarly apply $BO \perp AC$

50. Orthocentre =(5,9)
Altitude through C which is perpendicular to $\overline{AB}$ is 3x-2y+3=0

51. Apply $L_1\lambda_1 = L_2\lambda_2 = L_3\lambda_3$

52. $a_1a_2+b_1b_2=-32<0,\ c_1c_2=130>0$
Use $\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$

53. Bisector not containg origin is
$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=-\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$
$(\because c_1,c_2\ have\ opposite\ signs)$

54. $(l^2+m^2)(ax+bx+c)=2(al+bm)(lx+my+n)$

55. Diagonals are parallel to angular bisectors.

56. T divides PR in the ratio PQ:QR=1:6

57. Slope of reflected ray is $\frac{1}{\sqrt{3}}$ and it passing through
$(\sqrt{3},0)$ is $\frac{y-0}{x-\sqrt{3}}=\frac{1}{\sqrt{3}}$ $\sqrt{3}y = x-\sqrt{3}$

58. P.I of given line and AB

59. Equation of the straight line having equal intercepts is x+y = k and proceed.

60. Equate the distance from $(0,0)$ to the line $\frac{x}{x_1}+\frac{y}{y_1}=2$

61. Verification

62. AB<d

63. Given lines are concurrent.

LEVEL-II (H.W)

Slope of a line:

1. If the inclination of the line (2-k) x-(1-k) y+ (5-2k) = 0 is $\frac{3\pi}{4}$ then the value of k is

1) $\frac{5}{2}$ 2) $\frac{-3}{2}$ 3) $\frac{2}{3}$ 4) $\frac{3}{2}$

2. If the line joining the points $(at_1^2, 2at_1)$ $(at_2^2, 2at_2)$ is parallel to y=x then t_1+t_2=

1) $\frac{1}{2}$ 2) 4 3) $\frac{1}{4}$ 4) 2

Slope-intercept form,slope-point form and two-point form:

3. The line 2x+3y+12=0 cuts the axes at A & B. Then the equation of the

perpendicular bisector of $\overline{AB}$ is
1) 3x-2y+5=0 2) 3x-2y+7=0
3) 3x-2y+9=0 4) 3x-2y+8=0

4. If t_1, t_2 are roots of the equation $t^2+\lambda t+1=0$ where λ is an arbitary constant, then the line joining the points $(at_1^2, 2at_1)$ $(at_2^2, 2at_2)$ always passes through the fixed point
1) (-a, 0) 2) (0,a) 3) (a,0) 4) (0,-a)

5. ABCD is a parallelogram. Equations of AB and AD are 4x + 5y = 0 and 7x + 2y = 0 and the equation of diagonal BD is 11x + 7y + 9=0. The equation of AC is
1) $x+y=0$ **2)** $x-y=0$
3) $x+y+1=0$ 4) $x+y-1=0$

Intercepts and intercept form:

6. The equation of the straight line whose intercepts on x-axis and y-axis are respectively twice and thrice of those by the line 3x + 4y = 12, is
1) 9x + 8y = 72 2) 9x - 8y = 72
3) 8x + 9y = 72 4) 8x+9y+72=0

7. The equation of a straight line parallel to 2x + 3y + 11 = 0 and which is such that the sum of its intercepts on the axes is 15.
1) 2x+3y=15 2) 3x+2y=10
3) 2x-3y=10 4) 2x+3y=18

8. The straight line through P(1, 2) is such that its intercept between the axes is bisected at P. Its equation is
1) x + 2y = 5 2) x + y – 3 = 0
3) x – y + 1 = 0 4) 2x + y – 4 = 0

9. If (4, –3) divides the line segment between the axes in the ratio 4 : 5 then its equation is
1) 15x + 16y – 12 = 0 2) 3x – 4y – 24 = 0
3) 15x – 16y +108 = 0 4) 15x – 16y – 108 = 0

Normal form and symmetric form:

10. If a line AB makes an angle θ with OX and is at a distance of p units from the origin then the equation of AB is
1) xsin θ – ycos θ = p
2) xcos θ + ysin θ = p
3) xsin θ + ycos θ = p
4) xcos θ – ysin θ = p

11. The parametric equation of a line is given by $x=-2+\frac{r}{\sqrt{10}}$ and $y=1+3\frac{r}{\sqrt{10}}$
Then, for the line
1) intercept on the $x-axis=\frac{7}{3}$
2) intercept on the $y-axis=-7$
3) slope of the line =1/3
4) slope of the line =3

Problems on distances:

12. A straight line through the origin 'O' meets the parallel lines $4x+2y=9$ and $2x+y+6=0$ at points P and Q respectively, then point O divides the segment PQ in the ratio
1) 1 : 2 2) 3 : 4 3) 2 : 1 4) 4 : 3

13. The lengths of the perpendiculars from (m^2, 2m), (mn, m + n) and (n^2, 2n) to the straight line
x cos α + y sin α + sin α tan α = 0 are in
1) A.P. 2) G.P. 3) H.P. 4) A.G.P.

14. The distance between the Straight lines $y=mx+c_1$, $y=mx+c_2$ is $|c_1-c_2|$ then m =
1)0 2)1 3)2 4)3

15. Distance between parallel lines 4x+6y+8=0, 6x+9y+15=0 is
1) $2/\sqrt{13}$ 2) $1/\sqrt{13}$ 3) $3/\sqrt{13}$ 4) $4/\sqrt{13}$

16. 2x + 3y – 5 = 0, 2x + 3y + 15 = 0, x + y – 7 = 0, x + y + 7 = 0 are sides of a parallelogram. Then the centre of the parallelogram is
1) (–5,–5) 2) (5,–5) 3) (–5,5) 4) (5,5)

17. The distance of the point (3, 5) from the line 2x + 3y – 14 = 0 measured parllel to the line
x – 2y = 1 is
1) $\frac{7}{\sqrt{5}}$ 2) $\frac{7}{\sqrt{13}}$ 3) $\sqrt{5}$ 4) $\sqrt{13}$

18. Equation of the straight line passing through (1,1) and at a distance of 3 units from (–2, 3) is
1) $x-2=0$ 2) $5x-12y+6=0$
3) $5x-12y+7=0$ 4) $y-1=0$

Position of a point (s) w.r.to line (s):

19. If L_1,L_2 denote the lines x + 2y – 2 = 0,

$2x + 3y + 4 = 0$

1. L_1 is nearer to origin than L_2
2. L_2 is nearer to origin than L_1
3. L_1, L_2 are equidistant from origin 4. can't say

20. If the point (a, a) falls between the lines $|x+y|=2$, then:

1) $|a|=2$ 2) $|a|=1$ 3) $|a|<1$ 4) $|a|<\frac{1}{2}$

21. If $(2a-3, a^2-1)$ is on the same side of the line $x + y - 4 = 0$ as that of origin then the set of values of 'a' is

1) (-4,2) 2) (-2,4) 3)(-7,8) 4)(-7,5)

22. The range of values of the ordinate of a point moving on the line x=1 and allways remaining in the interior of the triangle formed by the lines y = x, the x-axis and x+y=4

1) (0,1) 2) (0,2) 3) (1,2) 4) (2,1)

Point of intersection of lines and concurrency of Lines :

23. If the line $\frac{x-1}{2}=\frac{y-2}{3}=t$ intersects the line x+y=8 then t =

1) 1 2) 2 3) 3 4) 4

24. Equation of line which is equally inclined to the axis and passes through a common points of family of lines $4acx + y(ab + bc + ca - abc) + abc = 0$ (where a, b, c > 0 are in H.P.) is

1) $y-x=\frac{7}{4}$ 2) $y+x=\frac{7}{4}$

3) $y-x=\frac{1}{4}$ 4) $y+x=\frac{3}{4}$

25. If a,b,c in GP then the line $a^2x+b^2y+ac=0$ always passes through the fixed point

1) (0, 1) 2) (1, 0) 3) (0, -1) 4) (1, -1)

26. If $U \equiv x+y-2=0$, $V \equiv 2x-3y+1=0$, the point of intersection of the lines 50U+7V=0, 3U+11V=0 is

1) (0,0) 2) (1,0) 3) (0,1) 4) (1,1)

27. The straight lines x+2y-9=0, 3x+5y-5=0 and ax+by-1 are concurrent if the straight line 22x-35y-1=0 passes through the point

1) (a, b) 2) (b,a) 3) (-a,b) 4) (-a, -b)

28. The equation of the line passing through the point of intersection of the lines 2x+y=5 and y=3x-5 and which is at the minimum distance from the point (1,2) is

1) x+y=3 2) x-y=1

3) x-2y=0 4) 2x+5y=7

29. Given a family of lines $a(2x + y + 4) + b(x - 2y - 3) = 0$. The number of lines belonging to the family at a distance $\sqrt{10}$ from P(2, –3) is

1)0 2)1 3)2 4)4

Angle between lines:

30. The acute angle between the lines $lx + my = l+m$, $l(x-y) + m(x+y) = 2m$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3}$

31. The angle between the lines $x\cos\alpha + y\sin\alpha = p_1$ and $x\cos\beta + y\sin\beta = p_2$ where $\alpha > \beta$ is

1) $\alpha+\beta$ 2) $\alpha-\beta$ 3) $\alpha\beta$ 4) $2\alpha-\beta$

32. One vertex of an equilateral triangle is (2,3) and the equation of one side is x-y+5=0. Then the equations to other sides are

1) $y-3=-(2\pm\sqrt{3})(x-2)$
2) $y-3=(\sqrt{2}\pm1)(x-2)$
3) $y-3=(\sqrt{3}\pm1)(x-2)$
4) $y-3=(\sqrt{5}\pm1)(x-2)$

33. Two equal sides of an isosceles triangle are given by the equations 7x–y+3=0 and x+y–3=0. The slope of the third side is

1) $-3,\frac{1}{3}$ 2) $3,-\frac{1}{3}$ 3) $3,\frac{1}{3}$ 4) $-3,-\frac{1}{3}$

34. Let there are two lines $2x+3y+\lambda=0$ and $\lambda x-3y-1=0$. If the origin lies in the obtuse angle then

1) $\lambda = \frac{9}{2}$ 2) $-2 < \lambda < 0$

3) $0 < \lambda < \frac{9}{2}$ 4) None of these

Triangles,area of the triangle:

35. **The area of the triangle formed by the axes and the line (cosh α - sinh α) x+ (cosh α + sinh α) y=2 in square units is**

1) 4 2) 3 3) 2 4) 1

36. **The equation to the base of an equilateral triangle is $\left(\sqrt{3}+1\right)x+\left(\sqrt{3}-1\right)y+2\sqrt{3}=0$ and opposite vertex is A(1,1) then the Area of the triangle is**

1) $3\sqrt{2}$ 2) $3\sqrt{3}$ 3) $2\sqrt{3}$ 4) $4\sqrt{3}$

37. **Equation of the line on which the perpendicular from the origin makes an angle of 30° with X-axis and which forms a triangle of area $\frac{50}{\sqrt{3}}$ with the axes is**

1) $\sqrt{2}x+2y=9$ 2) $\sqrt{2}x+3y=\pm 9$

3) $\sqrt{3}x-y=\pm 10$ 4) $\sqrt{3}x+y=\pm 10$

Quadrilaterals and area of the quadrilaterals:

38. **The point (2,3) is reflected four times about co-ordinate axes continuously starting with x-axis. The area of quadrilateral formed in sq.units is**

1) 24 2) 6 3) 12 4) 5

39. **Area of the quadrilateral formed by the lines $4y-3x-a=0$, $3y-4x+a=0$, $4y-3x-3a=0$, $3y-4x+2a=0$ is**

1) $\frac{a^2}{5}$ 2) $\frac{a^2}{7}$ 3) $\frac{2a^2}{7}$ 4) $\frac{2a^2}{9}$

40. **Two sides of a rectangle are 3x+4y+5-0, 4x-3y+15=0 and its one vertex is (0,0). Then the area of the rectangle is**

1) 4 2) 3 3) 2 4) 1

41. **The area enclosed with in the curve |x|+|y|=1 is**

1) 1 2) 2 3) $2\sqrt{2}$ 4) 4

Foot and Image:

42. **Foot of the perpendicular of $(6,8)$ in the line $x = y$ is**

1) (6,6) 2) (7,7) 3) (–6,–6) 4) (–7,–7)

43. **P is the midpoint of the part of the line 3x+y-2=0 intercepted between the axes. Then the image of P in origin is**

1) $\left(-1,-\frac{1}{3}\right)$ 2) $\left(-\frac{1}{3},-4\right)$

3) $\left(-\frac{1}{3},-1\right)$ 4) (-2, -3)

44. **The image of the point P (3,5) with respect to the line $y=x$ is the point Q and the image of Q with respect to the line y = 0 is the point R (a,b), then (a, b)=**

1) (5,3) 2) (5,-3)

3) (-5,3) 4) (-5,-3)

45.. **The equation of perpendicular bisector ofr AB and AC of a triangle ABC are x-y-5=0 and** $x+2y=0$ respectively. If A=(1,-2) then the **equation of $\overline{BC}$ is**

1) 14x+2y-41=0 2) 11x+2y-25=0

3) 2x-y-10=0 4) 14x-23y+40=0

Centroid, circumcentre, orthocentre and incentre:

46. **Let $O(0,0), P(3,4), Q(6,0)$ be the vertices of the triangle OPQ.The point R inside the triangles OPQ is such that the traingles OPR,PQR. OQR are of equal area. The coordinates of R are**

1) $\left(\frac{4}{3},3\right)$ 2) $\left(3,\frac{2}{3}\right)$

3) $\left(3,\frac{4}{3}\right)$ 4) $\left(\frac{4}{3},\frac{2}{3}\right)$

47. **If the circumcentre of the triangle lies at (0,0) and centroid is mid point of the line joining the points (2,3) and (4,7),then its orthocentre lies onthe line**

1)5x-3y=0 2)5x-3y+6=0

3)5x+3y=0 4)5x+3y+6=0

48. **The orthocentre of the triangle formed by the lines x+y=6, 2x+y=4 and x+2y=5 is**

1) $(10,-11)$ 2) $(-10,11)$
3) $(11,-10)$ 4) $(-11,-10)$

49. **The equation $x+2y=3$ represents the side BC of ΔABC; where co-ordinates of A are $(1,2)$. If the x-coordinate of the orthocentre of ΔABC is 3 then the y-coordinates of the orthocentre is :**
1) 4 2) 6 3) 8 4) 10

50. **The vertices A,B of a triangle are (2, 5), (4, -11). If C moves on the line $L \equiv 9x+7y+4=0$, then the locus of centroid of triangle ABC is parallel to**
1) AB 2) AC 3) BC 4) L

51. **Two sides of a triangle are $y = m_1 x$ and $y = m_2x$; m_1, m_2 are the roots of the equation $x^2 + ax - 1 = 0$. For all values of 'a' the orthocentre of the triangle lies at**
1) (1, 1) 2) (2, 2) 3) $\left(\frac{3}{2},\frac{3}{2}\right)$ 4) (0,0)

Angular bisectors :

52. **Equation of the line equidistant from the lines 2x+y+4=0, 3x+6y-5=0 is**
1) 3x-3y+17=0 2) 5x+7y-5=0
3) 3x-3y+19=0 4) 9x-9y+17=0

53. **Find the equation of the bisector of the angle between the lines x+2y–11=0, 3x–6y–5=0 which contains the point (1,–3).**
1) $2x-19=0$ 2) $2x+19=0$
3) $3x-19=0$ 4) $3x+19=0$

54. **The line 3x-3y+17=0 bisects the angle between a pair of lines of which one line is 2x+y+4=0, then the equation to the other line is**
1) 3x+6y-5=0 2) 3x+6y-7=0
3) 7x+14y=0 4) 4x-y+3=0

55. **The equation of a straight line passing through the point (4,5) and equally inclined to the lines 3x=4y+7 and 5y=12x+6 is**
1) 9x-7y=1 2) 9x+7y=1
3) 7x-9y=1 4) 7x-9 y=17

56. **If 2x+y-4=0 is bisector of the angle between the lines a(x–1)+b(y–2)=0, c(x–1)+d(y–2)=0, then the other bisector is**
1. $x-2y+1=0$ 2. $x-2y-3=0$
3. $x-2y+3=0$ 4. $x-2y-5=0$

Optimization and reflection in surface:

57. **A ray of light passing through the point (8,3) and is reflected at (14,0) on x axis. Then the equation of the reflected ray**
1) x+y=14 2) x-y=14
3) 2y=x-14 4) 3y=x-14

58. **Let P(1,1) and Q(3,2) be given points. The point R on the x-axis such that PR+RQ is minimum is**
1) $\left(\frac{5}{3},0\right)$ 2) $(2,0)$ 3) $(3,0)$ 4) $\left(\frac{3}{2},0\right)$

Miscellaneous problems:

59. **The vertices of triangle are $A(m,n), B(12,19)$ and $C(23,20)$, where m and n are integer. If its area is 70 and the slope of the median through A is –5, then m+n is**
1) 47 2) 27 3) 107 4) 43

60. **Number of circles touching the lines 3x+4y–1=0, 4x–5y+2=0 and 6x+8y+3=0 is**
1) 0 2) 2 3) 4 4) infinite

61. **The pont on the line $3x+4y=5$ which is equidistance from (1,2) and (3,4) is (EAM 2009)**
1) $(7,-4)$ 2) $(15,-10)$ 3) $\left(\frac{1}{7},\frac{8}{7}\right)$ 4) $\left(0,\frac{5}{4}\right)$

62. **The number of points p(x,y) with natural numbers as coordinates that lie inside the quadrilateral formed by the lines $2x+y=2,\ x=0, y=0$ and $x+y=5$ is (EAM- 2011)**
1) 12 2) 10 3) 6 4) 4

63. **A point moves in the xy plane such that the sum of its distance from two mutually perpendicular lines is always equal to 5 units.The area (in square units) enclosed by the locus of the point (EAM- 2012)**
1) $\frac{25}{4}$ 2) 25 3) 50 4) 100

LEVEL-II (H.W)- KEY

1) 4	2) 4	3) 1	4) 1	5) 2
6) 1	7) 4	8) 4	9) 4	10) 1
11) 4	12) 2	13) 2	14) 1	15) 2
16) 2	17) 3	18) 3	19) 1	20) 3
21) 1	22) 1	23) 1	24) 1	25) 3

26) 4	**27) 2**	**28) 1**	**29)2**	**30) 1**
31) 2	**32) 1**	**33) 1**	**34) 3**	**35) 3**
36) 3	**37) 4**	**38) 1**	**39) 3**	**40) 2**
41) 2	**42) 2**	**43) 3**	**44) 2**	**45) 2**
46) 3	**47) 1**	**48) 4**	**49) 2**	**50) 4**
51) 4	**52) 1**	**53) 3**	**54) 1**	**55) 1**
56) 3	**57) 3**	**58) 1**	**59) 3**	**60) 2**
61) 2	**62) 3**	**63) 3**		

LEVEL-II (H.W)-HINTS

1. $\tan\frac{3\pi}{4}=\frac{2-k}{1-k}$

2. Slope of the parllel lines are equal

3. $A(-6,0)\,B(0,-4)$ perpendicular bisector of AB is $y+2=\frac{3}{2}(x+3)$

4. Equation of the line is $y(t_1+t_2)-2at_1(t_1+t_2)$ $=2x-2at_1^{\,2}$

5. by solving AB,BD we get B(-5/3, 4/3)
by solving AD,BD we get D(2/3, -7/3)
mid point of B.D lies on AC

6. $a=8, b=6$

7. sum of the intercepts of the line 2x+3y+k=0 is 15.
$\frac{-k}{2}+\frac{-k}{3}=15$

8. $\left(\frac{a}{2},\frac{b}{2}\right)=(1,2)$

9. $\frac{nx}{x_1}+\frac{my}{y_1}=m+n$

10. The line makes an angle θ with x-axis then its perpendicular makes an angle $\alpha=\theta-90^0$

11. Point $(-2,1)$ slope $=3$

12. $\therefore\left|\frac{OP}{OQ}\right|=\left|\frac{9/\sqrt{20}}{-6/\sqrt{5}}\right|=\left|\frac{9}{12}\right|=\frac{3}{4}=3:4$

13. Put $\alpha=0$ then given equation of line is x=0

14. $\frac{|c_1-c_2|}{\sqrt{1+m^2}}=|c_1-c_2|$

15. multiply first equation by 3 and second equation by 2

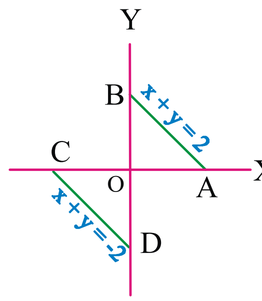
16. Centre lies on a line parallel to given lines and mid way between them

17. $\left|\frac{ax_1+by_1+c}{a\cos\theta+b\sin\theta}\right|$ where $\tan\theta=\frac{1}{2}$

18. verification

19. verification

20. From the figure
$-1<a<1$ *i.e.* $|a|<1$.



21. since $L_0<0$ we have $L_A<0$

22. From figure 0 < ordinate of P<1

23. Point (2t+1,3t+2) lies on x+y=8

24. Lines can be written
$\frac{4}{b}x+y\left(\frac{3}{b}\right)+1-y=0,\ \frac{1}{b}(4x+3y)+1-y=0$
$\Rightarrow$ Lines are concurrent at $\left(-\frac{3}{4},1\right)$
$\therefore$ Required line is $y-1=\pm1\left(x+\frac{3}{4}\right)$

25. Given equation is $a^2x+b^2(y+1)=0$

26. Point of intersection of u & v

27. $\begin{vmatrix}1 & 2 & -9\\ 3 & 5 & -5\\ a & b & -1\end{vmatrix}=0$

28. The point $(1,2)$ lies on $L_1+\lambda L_2=0$

29. $P=\left|\frac{a(4-3+4)+b(2+6-3)}{\sqrt{(2a+b)^2+(a-2b)^2}}\right|=\sqrt{10}$
$25\,(a+b)^2=10\,(5a^2+5b^2)$
$25\,(a-b)^2=0$
$a=b$

30. $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right|$

31. $\cos\theta = \dfrac{a_1 a_2 + b_1 b_2}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}$

32. apply $\dfrac{m \pm \tan\alpha}{1 \mp m\tan\alpha}$

33. Apply $\dfrac{m_1 - m}{1 + m_1 m} = -\dfrac{m_2 - m}{1 + m_2 m}$

34. Origin will lie in obtuse angle if $cc'(aa' + bb') > 0$

$\Rightarrow -\lambda(2\lambda - 3\times 3) > 0 \Rightarrow -\lambda(2\lambda - 9) > 0$

$\Rightarrow 9\lambda - 2\lambda^2 > 0 \qquad \lambda \in \left(0, \dfrac{9}{2}\right)$

35. $\Delta = \dfrac{c^2}{2|ab|}$

36. Area of an equilateral triangle is

$\dfrac{h^2}{\sqrt{3}}$ where h is the height of the triangle

37. Equation of the line is $x\cos 30^0 + y\sin 30^0 = P$

38. $A(2,3)B(-2,3)C(-2-3)D(2,-3)$

39. The area of the parallelogram formed by the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + d_1 = 0$, $a_1x + b_1y + c_2 = 0$, $a_2x + b_2y + d_2 = 0$ is $\left|\dfrac{(c_1 - c_2)(d_1 - d_2)}{a_1b_2 - a_2b_1}\right|$ Sq.units.

40. The perpendicular distance from orign lines are a,b. Then area = ab

41. From the diagram required area $= 4(area\ of\ \Delta) = 2$

42. Use foot of the perpendicular formula

43. Midpoint of intercepts is $\left(\dfrac{1}{3}, 1\right)$

44. Image of P(3,5) w. r to the line y = x is Q(5,3)

45. B,C are images of A w.r.to given lines

46. Centroid ΔOPQ

47. circumcentre,centroid and orthocentre lies on asame line.

48. Use formula given in synopsis

49. $AD \perp BC$

$\Rightarrow \left[\dfrac{\lambda - 2}{3 - 1}\right]\left[-\dfrac{1}{2}\right] = -1$

or $2 - \lambda = -4$

$-\lambda = -6 \Rightarrow \lambda = 6$.

50. Choose $C = (\alpha, \beta)$

$G(x_1, y_1) = \left(\dfrac{\alpha + 6}{3}, \dfrac{\beta - 6}{3}\right)$

$\Rightarrow \alpha = 3x_1 - 6, \beta = 3y_1 + 6$

Substance (α, β) lies on L=0

51. given triangle is right angled at origin

52. Angular bisector

53. Using formula given in synopsis

54. Verify angular bisector formula

55. $(x_1, y_1) = (4,5)$

Slope of angle of bisector are $\dfrac{9}{7}, \dfrac{-7}{9}$

56. required bisector is perpendicular to given and passes through (1,2)

57. Write the image of $(8,3)$ in X-axis and write the equation through that point and $(14,0)$

58. Image of P in x-axis is $P^1 = (1,-1)$, R is intersection of x-axis and line QP^1

59. $Area = 70 \Rightarrow -m + 11n = 337$

$median\ slope = -5 \Rightarrow 5m + n = 107$

60. Two lines are parallel

61. Required point $p(x_1, y_1)$ lies on given line then $3x_1 + 4y_1 = 5$ and PA=PB$\Rightarrow x_1 + y_1 = 5$

62. Draw the diagram and observe the points $(1,1),(1,2),(2,1),(2,2),(1,3)$ and $(3,1)$ are inside of quadrilateral.

63. From given data $|x| + |y| = 5$ hence required

area $= \dfrac{2(5)^2}{|(1)(1)|} = 50$

LEVEL-III

1. The line 3x-2y = 24 meets x-axis at A and y-axis at B. The perpendicular bisector of AB meets the line through (0, -1) and parallel to x-axis at C. Then C is

1) $\left(\frac{-7}{2},-1\right)$ 2) $\left(\frac{-15}{2},-1\right)$

3) $\left(\frac{-11}{2},-1\right)$ 4) $\left(\frac{-13}{2},-1\right)$

2. A square of side "a" lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle α where $0<\alpha<\frac{\pi}{4}$ with the positive direction of x - axis. The equation of its diagonal not passing through the origin is

1) $y(\cos\alpha+\sin\alpha)+x(\cos\alpha-\sin\alpha)=a$

2) $y(\cos\alpha-\sin\alpha)-x(\sin\alpha-\cos\alpha)=a$

3) $y(\cos\alpha+\sin\alpha)+x(\sin\alpha-\cos\alpha)=a$

4) $y(\cos\alpha+\sin\alpha)+x(\cos\alpha+\sin\alpha)=a$

3. A particle is moving in a straight line and at some moment it occupied the positions (5,2) and (-1,-2). Then the position of the particle when it is on x-axis is

1) (-2, 0) 2) (0, 2) 3) (2, 0) 4) (4, 0)

4. If PS is the median of the triangle with vertices P(2,2), Q(6,-1) and R(7,3) then the equation of the line passing through (1,-1) and parallel to $\overline{PS}$ (AIEEE 2014)

1) 4x-7y-11=0 2) 2x+9y+7=0
3) 4x+7y+3=0 4) 2x-9y-11=0

5. The Point P(2 , 1) is shifted by $3\sqrt{2}$ parallel to the line $x+y=1$, in the direction of increasing ordinate, to reach Q. The image of Q by the line $x+y=1$ is

1) $(5,-2)$ 2) $(-1,4)$ 3) $(3,-4)$ 4) $(-3, 2)$

6. Distance of origin from line $\left(1+\sqrt{3}\right)y+\left(1-\sqrt{3}\right)x=10$ along the line $y=\sqrt{3}x+k$ is

1) $5/\sqrt{2}$ 2) $5/\sqrt{2}+k$ 3) 10 4) 5

7. One of the diagonals of a square is the portion of the line $\frac{x}{2}+\frac{y}{3}=2$ intercepted between the axes. Then the extremities of the other diagonal are

1) $(5,5),(-1,1)$ 2) $(0,0),(4,6)$
3) $(0,0),(-1,1)$ 4) $(5,5),(4,6)$

8. If the line $y=\sqrt{3}x$ cuts the curve $x^3+y^3+3xy+5x^2+3y^2+4x+5y-1=0$ at the points A,B,C then OA.OB.OC is

1) $\frac{4}{13}\left(3\sqrt{3}-1\right)$ 2) $3\sqrt{3}+1$

3) $\frac{2}{\sqrt{3}}+7$ 4) $3\sqrt{3}-1$

9. Each side of a square is of length 4. The centre of the square is (3, 7) and one of its diagonals is parallel to y=x. Then co-ordinates of its vertices are

1) (1,5), (1,9), (5,9), (5,5)
2) (2,5), (2,7), (4,7), (4,4)
3) (2,5), (2,6), (3,5), (3,6)
4) (5,2), (6,2), (5,3), (6,3)

10. If the line $y-\sqrt{3}x+3=0$ cuts the curve $y^2=x+2$ at A and B and point on the line P is $\left(\sqrt{3},0\right)$ then $|PA.PB|=$

1) $\frac{4\left(\sqrt{3}+2\right)}{3}$ 2) $\frac{4\left(2-\sqrt{3}\right)}{3}$

3) $\frac{4\sqrt{3}}{2}$ 4) $\frac{2\left(\sqrt{3}+2\right)}{3}$

11. Points A and B are in the first quadrant; point 'O' is the origin. If the slope of OA is 1, slope of OB is 7 and $OA=OB$, then the slope of AB is

1) $-\frac{1}{5}$ 2) $-\frac{1}{4}$ 3) $-\frac{1}{3}$ 4) $-\frac{1}{2}$

12. The line joining the points A(3,0) and B(5,2) is rotated about A in the anticlockwise direction through an angle of 15^0. If B goes to C in the new position now the line joining A and C is rotated about A in the

anticlockwise direction through an angle of 45^0 of C goes to D in the new position, then the coordinates of D are

1) $\left(4-\sqrt{3},\sqrt{3}-1\right)$ 2) $\left(4+\sqrt{3},\sqrt{3}-1\right)$

3) $\left(4-\sqrt{3},\sqrt{3}+1\right)$ 4) $\left(4+\sqrt{3},\sqrt{3}+1\right)$

13. **The equation of the line passing through (1,2) and having a distance equal to 7 units from the point (8,9) is**

1) $y=3x-1$ 2) $y=2$

3) $x=1$ 4) $x+y=3$

14. **Find the values of non-negative real numbers h_1,h_2,h_3,k_1,k_2,k_3 such that the algebraic sum of the perpendiculars drawn from points $(2,k_1)$, $(3,k_2)$, $(7,k_3)$, $(h_1,4)$, $(h_2,5)$, $(h_3,-3)$ on a variable line passing through $(2,1)$ is zero.**

1) $h_1=h_2=h_3=k_1=k_2=k_3=0$

2) $h_1=h_2=h_3=k_1=k_2=k_3=1$

3) $h_1=h_2=h_3=k_1=k_2=k_3=2$

4) $h_1=h_2=h_3=k_1=k_2=k_3=4$

15. **Three lines $x+2y+3=0$; $x+2y-7=0$ and $2x-y-4=0$ form the three sides of two squares. The equation to the fourth side of each square is**

1) $2x-y+14=0$ *and* $2x-y+6=0$

2) $2x-y+14=0$ *and* $2x-y-6=0$

3) $2x-y-14=0$ *and* $2x-y-6=0$

4) $2x-y-14=0$ *and* $2x-y+6=0$

16. **The line L given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point $(13,32)$. The line K is parallel to L and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then the distance between L and K is** **[AIEEE-2010]**

1) $\sqrt{17}$ 2) $\frac{17}{\sqrt{15}}$ 3) $\frac{23}{\sqrt{17}}$ 4) $\frac{23}{\sqrt{15}}$

17. **If the line $2x+y=k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals** **[AIEEE - 2012]**

1) $\frac{29}{5}$ 2) 5 3) 6 4) $\frac{11}{5}$

18. **If the point $P\left(a^2,a\right)$ lies in the region corresponding to the acute angle between the lines $2y=x$ and $4y=x$, then**

1) $a\in[2,4]$ 2) $a\in(2,4]$

3) $a\in[2,5)$ 4) $a\in(2,4)$

19. **The set of values of 'b' for which the origin and the point (1,1) lie on the same side of the straight line $a^2x+aby+1=0,\ \forall a\in R, b>0$ are**

1) $b\in(2,4)$ 2) $b\in(0,2)$

3) $b\in[0,2]$ 4) $b\in[0,3]$

20. **The equations of sides of a triangle are 7x–5y–11=0, 8x+3y+31=0, x+8y–19=0. Then the point (0,0) lies**

1) inside of triangle 2) outside of triangle

3) on the triangle 4) can't say

21. **If the points of intersection of lines $L_1: y-m_1x-k=0$ and $L_2: y-m_2x-k=0\ (m_1\neq m_2)$ lies inside the triangle formed by the lines $2x+3y=1, x+2y=3$ and $5x-6y-1=0$, then true set of values of k are**

1) $\left(\frac{1}{3},\frac{3}{2}\right)$ 2) $\left(\frac{1}{2},1\right)$ 3) $\left(0,\frac{3}{2}\right)$ 4) $\left(\frac{-3}{2},0\right)$

22. **The range of value of α such that $(0,\alpha)$ lies on or inside the triangle formed by the lines y + 3x + 2 = 0, 3y –2x–5 = 0, 4y +x –14 = 0 is**

1) $5<\alpha\le 7$ 2) $\frac{1}{2}\le\alpha\le 1$.

3) $\frac{5}{3}\le\alpha\le\frac{7}{2}$ 4) $\frac{1}{3}\le\alpha\le\frac{1}{2}$

23. **The lines $x+y=|a|$, and $ax-y=1$ inter-**

sect each other in the first quadrant then the set of all possible values of a is the interval [AIEEE-2011]

1) $(0,\infty)$ 2) $(1,\infty)$ 3) $(-1,\infty)$ 4) $[-1,1]$

24. Let a , b ,c and d be non zero numbers . If the point of intersection of the lines $4ax+2ay+c=0$ and $5bx+2by+d=0$ lies in the fourth quadrant and is equidistance from the two axes then (MAINS 2014)

1) $2bc-3ad=0$ 2) $2bc+3ad=0$

3) $3bc-2ad=0$ 4) $3bc+2ad=0$

25. If the lines x+ay+a=0, bx+y+b=0, cx+cy+1=0 (a, b, c being distinct and $\neq 1$) are concurrent then the value of

$\frac{a}{a-1}+\frac{b}{b-1}+\frac{c}{c-1}=$

1) -1 2) 0 3) 1 4) 3

26. If $4a^2+9b^2-c^2+12ab=0$ then the family of straight lines $ax+by+c=0$ is concurrent at

1) (2,3) or (-2,-3) 2) (2,-3) or (-2,6)

3)(-2,-4) or (-2,3) 4) (2,5) or (-1,-5)

27. If $a^2 - b^2 - c^2 - 2bc = 0$, then the family of lines ax + by + c =0 are concurrent at the points

1) (1, –1) 2) (–1, 1) 3)(1, 0) 4)(–1, –1)

28. If $t_1 \neq t_2 \neq t_3$ and the lines $t_1x+y=2at_1+at_1^3$; $t_2x + y = 2at_2 + at_2^3$; $t_3x + y= 2at_3 + at_3^3$ are concurrent then $t_1 + t_2 + t_3$ is

1. 0 2. -1 3.1 4. 2

29. Consider the family of lines

$(x+y-1)+\lambda(2x+3y-5)=0$ and

$(3x+2y-4)+\mu(x+2y-6)=0$,

equation of a straight line that belongs to both the families is

1) $x-2y-8=0$ 2) $x-2y+8=0$

3) $2x+y-8=0$ 4) $2x-y-8=0$

30. If a,b and c are three consecutive odd integers then the variable line ax+by+c=0 always passes through

1) (2 1) 2) (1 2) 3) (-1 2) 4) (1 -2)

31. One vertex of an equilateral triangle is (2,3) and the equation of one side is x-y+5=0 then the equations to the other sides are

1) y-3 = $-\left(2\pm\sqrt{3}\right)$ (x-2)

2) y-3 = $\left(\sqrt{2}\pm 1\right)$ (x-2)

3) y-3 = $\left(\sqrt{3}\pm 1\right)$ (x-2)

4) y-3 = $\left(\sqrt{5}\pm 1\right)$ (x-2)

32. Let *P* (2, -4) and *Q* (3, 1) be two given points. Let *R* (x, y) be a point such that (x-2) (x-3) + (y-1) (y+4) = 0. If area of ΔPQR is $\frac{13}{2}$, then the number of possible positions of *R* are

1) 2 2) 3 3) 4 4) 6

33. If the base of an isosceles triangle is of length 2P and the length of the altitude dropped to the base is q, then the distance from the mid point of the base to the side of the triangle is

1) $\frac{pq}{\sqrt{p^2+q^2}}$ 2) $\frac{2pq}{\sqrt{p^2+q^2}}$

3) $\frac{3pq}{\sqrt{p^2+q^2}}$ 4) $\frac{4pq}{\sqrt{p^2+q^2}}$

34. If m_1 and m_2 are the roots of the equation $x^2-ax-a-1=0$, then the area of the triangle formed by the three straight lines $y=m_1x, y=m_2x\ and\ y=a(a\neq -1)$ is

1) $\frac{-a^2(a+2)}{2(a+1)} if\ a>-1$

2) $\frac{+a^2(a+2)}{2(a+1)} if\ a<-1$

3) $\frac{-a^2(a+2)}{2(a+1)} if -2<a<-1$

4) $\frac{a^2(a+2)}{2(a+1)} if\ a>-2$

35. The equation of a straight line L is x+y=2, and L_1 is another straight line perpendicular to L and passes through the piont (1/2, 0), then area of the triangle formed by the y-axis and the lines L, L_1 is

1) $\frac{25}{8}$ 2) $\frac{25}{16}$ 3) $\frac{25}{4}$ 4) $\frac{25}{12}$

36. In an isosceles triangle OAB, O is the origin and OA = OB = 6. The equation of the side AB is x-y+1=0. Then the area of the triangle is

1) $2\sqrt{21}$ 2) $\sqrt{142}$ 3) $\frac{\sqrt{142}}{2}$ 4) $\frac{\sqrt{71}}{2}$

37. An equilateral triangle is constructed between two parallel lines $\sqrt{3}x+y-6=0$ and $\sqrt{3}x+y+9=0$ with base on one and vertex on the other. Then the area of triangle is

1) $\frac{200}{\sqrt{3}}$ 2) $\frac{225}{4\sqrt{3}}$ 3) $\frac{225}{\sqrt{3}}$ 4) $\frac{200}{4\sqrt{3}}$

38. Area of triangle formed by the lines 2x + y – 3 = 0, x + 4y – 5 = 0 and 3x + 5y – 1 = 0 is

1) 15/2 2) 49/2 3) 27/56 4) 7/2

39. If $f(x+y)=f(x)f(y)$ for all x and y if $f(1)=2,$ then area enclosed by $3|x|+2|y|\le 8$ is

1) $f(5)sq.units$ 2) $f(6)sq.units$

3) $1/3f(6)sq.units$ 4) $f(4)sq.units$

40. Four sides of a quadrilateral are given by the equation $xy(x-2)(y-3)=0$, then the equation of the line parallel to $x-4y=0$ that divides the quadrilateral into two equal parts is

1) $x-4y+5=0$ 2) $x-4y-5=0$

3) $x-4y+1=0$ 4) $x-4y-1=0$

41. L_1 and L_2 are two intersecting lines and the angle between the image of L_1 w.r.t L_2 and that of L_2 w.r.t.L_1 is 45°. Then the angle between L_1 and L_2 is

1) 20° 2) 15° 3) 45° 4) 60°

42. L_1 and L_2 are two intersecting lines. If the image of L_1 w.r.t. L_2 and that of L_2 w.r.t.L_1 concide, then the angle between L_1 and L_2 is

1) 35° 2) 60° 3) 90° 4) 45°

43. For all values of θ all the lines represented by the equation $(2\cos\theta + 3\sin\theta)x + (3\cos\theta - 5\sin\theta)y - (5\cos\theta - 2\sin\theta) = 0$ passes through a fixed point then the reflection of that point with respect to the line x+y = $\sqrt{2}$ is

1) $\left(\sqrt{2}+1,\sqrt{2}+1\right)$ 2) $\left(\sqrt{2}-1,\sqrt{2}-1\right)$

3) $\left(\sqrt{3}-1,\sqrt{3}-1\right)$ 4) $\left(\sqrt{3}+1,\sqrt{3}+1\right)$

44. The conbined equation of straight lines that can be obtained by reflecting the lines $y=|x-2|$ in the y-axis is

1) $y^2+x^2+4x+4=0$

2) $y^2+x^2-4x+4=0$

3) $y^2-x^2+4x-4=0$

4) $y^2-x^2-4x-4=0$

45. In ΔABC, B=(0, 0), AB=2, $\angle ABC=\frac{\pi}{3}$ and the middle point of BC has the co-ordinates (2, 0). Then the centroid of triangle is

1) $\left(\frac{5}{3},\frac{1}{3}\right)$ 2) $\left(\frac{5}{3},\frac{1}{\sqrt{3}}\right)$

3) $\left(\frac{5}{\sqrt{3}},\frac{1}{3}\right)$ 4) $\left(\frac{5}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

46. In triangle ABC, co-ordinates of A are $(-1,3)$ and equation of medians and altitude through point B are $2x+y=8$ and $2x+3y=8$ respectively, then

1) coordinates of C are (4,0)

2) coordinates of C are (3,9)

3) coordinates of C are (3,3)

4) coordinates of centroid are (2,2)

47. The sides of a triangle are x+y=1, 7y = x and $\sqrt{3}$y + x = 0. Then the following is an interior point of the traingle

1) Circumcentre 2) Centroid

3) Orthocentre 4) Cannot say

48. If the equations of the sides of a trinagle are 2x + y = 2, y = x, $\sqrt{3}y+x=0$ then which of the following is an exterior point of triangle.

1) orthocentre 2) incentre
3) centroid 4) Cannot say

49. **One vertex of the equilateral triangle with centroid at the origin and one side as *x+y-2=0* is**
1) $(-2,-2)$ 2) $(2,2)$ 3) $(2,-2)$ 4) $(-2,2)$

50. **A ray of light is sent along the line x–2y–3=0. On reaching the line 3x–2y–5=0, the ray is reflected from it. The equation of the line containing the the reflected ray.**
1) $29x-2y+31=0$ 2) $29x+2y-31=0$
3) $29x-2y-31=0$ 4) $29x+2y+31=0$

51. **A light ray coming along the line $3x+4y=5$ gets reflected from the line $ax+by=1$ and goes along the line $5x-2y=10$. Then,**
1) $a=\frac{64}{115},\ b=\frac{112}{15}$ 2) $a=\frac{14}{15},\ b=\frac{-8}{115}$
3) $a=\frac{64}{115},\ b=\frac{-8}{115}$ 4) $a=\frac{64}{15},\ b=\frac{14}{15}$

52. **If x_1,y_1 are roots of $x^2+8x-20=0, x_2,y_2$ are the roots of $4x^2+32x-57=0$ and x_3,y_3 are the roots of $9x^2+72x-112=0$, then the points $(x_1,y_1),(x_2,y_2)$ *and* (x_3,y_3) where $x_i<y_i$ for i =1,2,3**
1) are collinear
2) form an equilateral triangle
3) form a right angled isosceles triangle
4) are concyclic

53. **Triangle is formed by the coordinates (0, 0), (0, 21) and (21, 0). The number of integral coordinates strictly inside triangle (integral coordinates has both *x* and *y* as integers) :**
1) 190 2) 105 3) 231 4) 205

54. **Origin is the centre of the square with one of its vertices at (3,4) then the other vertices are**
1) (-3, 4), (-3, -4), (3, -4)
2) (-4, 3), (-3, -4), (4, -3)
3) (-4, 3), (-4, -3), (3, -4)
4) (3, 4), (-4, -3), (4, -3)

55. **One side of a rectangle lies along the line 4x+7y+5=0. Two vertices are (-3,1), (1,1) then the remaining vertices are**
1) $\left(\frac{1}{65},\frac{-47}{65}\right),\left(\frac{-131}{65},\frac{177}{65}\right)$
2) $\left(\frac{-1}{65},\frac{47}{65}\right)\left(\frac{-131}{65},\frac{177}{65}\right)$
3) $\left(\frac{1}{65},\frac{-47}{65}\right)\left(\frac{131}{65},\frac{-177}{65}\right)$
4) (1, -47), (131, 47)

56. **All points lying inside the triangle formed by the points (1,3), (5,0), (-1,2) satisfy**
1) $2x+y-13=0$ 2) $3x+2y\geq 0$
3) $3x-4y-12\leq 0$ 4) $4x+y=0$

57. **If one vertex of an equilateral triangle of side a lies at the origin and the other lies on the line $x-\sqrt{3}y=0$, the coordinates of the third vertex are**
1) (0, –a) 2) (a, 0)
3) $\left(\frac{a\sqrt{3}}{2},\frac{a}{2}\right)$ 4) $\left(\frac{a\sqrt{3}}{2},-\frac{a}{2}\right)$

58. **let AB be a line segment of length 4 units with the point A on the line y=2x and B on the line y=x. Then the locus of middle point of all such line segment is**
1) a parabola 2) an ellipse
3) a hyperbola 4) a circle

LEVEL-III - KEY

1) 1	2) 1	3) 3	4) 2	5) 4
6) 4	7) 1	8) 1	9) 1	10) 3
11) 4	12) 2	13) 2	14) 1	15) 4

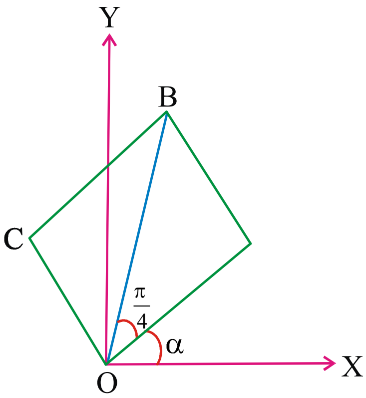
16) 3 17) 3 18) 4 19) 2 20) 1
21) 1 22) 3 23) 2 24) 3 25) 3
26) 1 27) 2 28) 1 29) 2 30) 4
31) 1 32) 1 33) 1 34) 3 35) 2
36) 4 37) 2 38) 4 39) 3 40) 1
41) 2 42) 2 43) 2 44) 4 45) 2
46) 2 47) 2 48) 1 49) 1 50) 3
51) 3 52) 1 53) 1 54) 2 55) 1
56) 2 57) 4 58) 2

LEVEL-III - HINTS

1. perpendicular bisector of AB is

$2x+3y+10=0.............(1)$

$y=-1.............(2)$,Solve the equations

2. 

Slope of BC $tan\left(\frac{\pi}{4}+\alpha\right)$

Slope of CA $-\cot\left(\frac{\pi}{4}+\alpha\right)$

Equation of CA

$(y-a\sin\alpha)=-\cot\left(\frac{\pi}{4}+\alpha\right)(x-a\cos\alpha)$

3. equation of line through given points is 2x-3y-4=0 when cuts x-axis at (2,0)
4. S=midpoint of Q,R = (13/2,1)
slope of $\overline{PS}$ = -2/9
5. $(x_1+r\cos\theta, y_1+r\sin\theta)=(-1,4)$
$r=3\sqrt{2}, \theta=3\pi/4$
6. The slope of the line is $\sqrt{3}$
Find the equation of the line passing through origin.Then find point of intersection

7. The equation of the other diagonal is

$$\frac{x-2}{\frac{3}{\sqrt{13}}}=\frac{y-3}{\frac{2}{\sqrt{13}}}=r$$

For the extremities of the diagonal,

$r=\pm\sqrt{13}$. Hence

$x-2=\pm 3, y-3=\pm 2$

$x=5,-1 \; and \; y=5,1$

Therefore, the extremities of the diagonal are (5,5) and (-1,1).

8. $\because y=\sqrt{3}x=x\tan 60^0$

$$\therefore \frac{x-0}{\cos 60^0}=\frac{y-0}{\sin 60^0}=r$$

$$\therefore (x,y)=\left(\frac{r}{2},\frac{r\sqrt{3}}{2}\right)$$

lies on

$x^3+y^3+3xy+5x^2+3y^2+4x+5y-1=0$

$$\therefore \frac{r^3}{8}+\frac{3\sqrt{3}r^3}{8}+\frac{3\sqrt{3}r^2}{4}+\frac{5r^2}{4}+\frac{9r^2}{4}+2r+\frac{5\sqrt{3}r}{2}-1=0$$

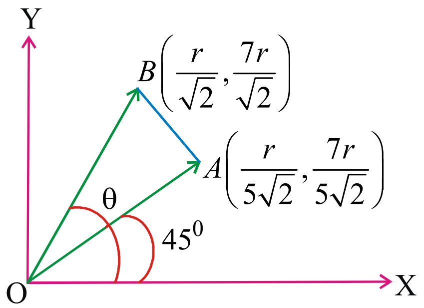
$$\Rightarrow\left(\frac{3\sqrt{3}+1}{8}\right)r^3+\left(\frac{14+3\sqrt{3}}{4}\right)r^2+\left(\frac{4+5\sqrt{3}}{2}\right)r-1=0$$

$$\therefore OA.OB.OC=r_1r_2r_3=\frac{1}{\left(\frac{3\sqrt{3}+1}{8}\right)}$$

9. Use $(x_1\pm r\cos\theta, y_1\pm r\sin\theta)$
$r=2\sqrt{2}$ And $\theta=45^0,135^0$

10. Let $PA=r$, then $A=\left(\sqrt{3}+\frac{r}{2},\frac{\sqrt{3}r}{2}\right)$.

11. $\tan\theta=7$
$OA=OB=r$
$\sin\theta=\frac{7}{5\sqrt{2}}, \cos\theta=\frac{1}{5\sqrt{2}}$

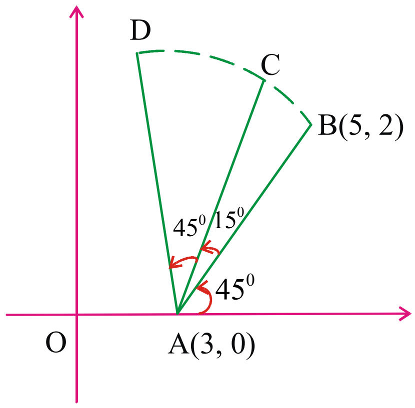


Now, $m_{AB}=-1/2$

12. Slope of $AB=1$

$r=AB=2\sqrt{2}$, $\theta=105^0$

$D=\left(x_1+r\cos\theta, y_1+r\sin\theta\right)=\left(4+\sqrt{3},\sqrt{3}-1\right)$



13. Distance from (8,9) to line $y-2=m(x-1)$ is $7\Rightarrow m=0$

14. Let the eqution of variable line be $ax+by+c=0$, it is given that

$$\sum_{i=1}^{6}\frac{ax_i+by_i+c}{\sqrt{a^2+b^2}}=0$$

$$\Rightarrow\ a\left(\frac{\Sigma x_i}{6}\right)+b\left(\frac{\Sigma y_i}{6}\right)+c=0$$

So, the fixed point must be $\frac{\Sigma x_i}{6},\frac{\Sigma y_i}{6}$. But fixed point is $(2,1)$ so

$(2+3+7+h_1+h_2+h_3)/6=2$

$\Rightarrow\ h_1+h_2+h_3=0$

$\Rightarrow\ h_1=0,\ h_2=0,\ h_3=0$

(as h_1,h_2,h_3 are non-negative)

similarly, we get

$\frac{k_1+k_2+k_3+4+5+-3}{6}=1\quad\Rightarrow$

$k_1=k_2=k_3=0$

15. Hence line is $2x-y+\lambda=0$

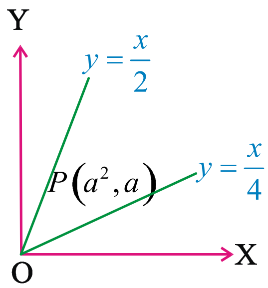
now $\left|\frac{\lambda+4}{\sqrt{5}}\right|=\frac{10}{\sqrt{5}}$

$\lambda+4=\pm10$

$\lambda=6\ or\ \lambda=-14\Rightarrow(B)$

16. Substitute the given point. Find 'b' and equate the slopes to find 'c' and apply distance between parallel lines.

17. Find ratio point and substitute in the line.

18. 

But $a^2>0$, hence point $P\left(a^2,a\right)$ lies in first quadrant.

We have $a-\left(\frac{a^2}{4}\right)>0$ and $a-\left(\frac{a^2}{2}\right)<0$

($\therefore$ $(1,0)$ and P lies on same side of $x-2y=0$ and $(1,0)$ and P lies opposite sides of $x-4y=0$)

$\Rightarrow\ 0<a$ and $a\in(-\infty,0)\cup(2,\infty)$

$\Rightarrow\ a\in(2,4)$

19. $D<0\Rightarrow b^2-4<0\Rightarrow -2<b<2\ but\ b>0$

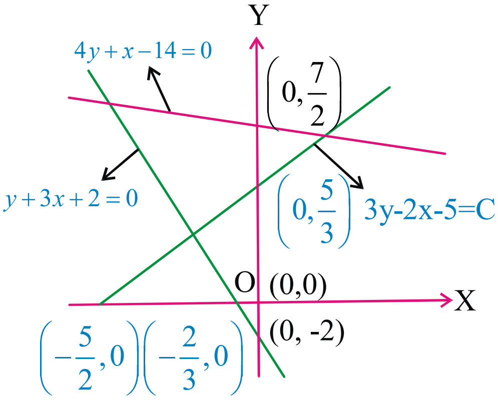
$\therefore\quad 0<b<2$

ie., $\quad b\in(0,2)$

20. Draw the diagram

21. Clearly point of intersection of L_1 and L_2, is (0,k) which lies on y - axis.

22.



from the figure, it is clear that $\frac{5}{3}\le\alpha\le\frac{7}{2}$

23. Point of intersection $=\left(\frac{1+|a|}{1+a},\frac{a|a|-1}{a+1}\right)$

since x>0, y>0 we have a+1>0 and $a|a|>1$

as $|a|\ge 0$, $a|a|>1$ we get a>0 thus $a^2>1$

or $a>1$

24. Point $(k,-k)$ satisfies both lines then

$c=-2ak, d=-3bk \Rightarrow \frac{c}{-2a}=\frac{d}{-3b}$

25. $\begin{vmatrix}1 & a & a\\ b & 1 & b\\ c & c & 1\end{vmatrix}=0$

Use det properties

26. $(2a+3b)^2-c^2=0$

27. $a^2-(b+c)^2=0$

$(a-b-c)(a+b+c)=0$

Either $a-b-c=0$ or $a+b+c=0$

$\Rightarrow$ $-a+b+c=0$

$\Rightarrow$ Family of lines passing through (–1, 1) and (1, 1).

28. t_1,t_2,t_3 roots of $y+xt=2at+at^3$

29. The family of lines

$(x+y-1)+\lambda(2x+3y-5)=0$ passes through a point such that

$x+y-1=0$, $2x+3y-5=0$

30. a,b,c are in A.P

31. Let slope of another side is m

$\tan 60^0=\left|\frac{m-1}{1+m}\right|$; $m=-2\pm\sqrt{3}$

32. ΔPQR is right angle triangle

radius = altitude

33. Consider $B(-p,0)C(p,0)$ $A(0,q)$

34. Since m_1,m_2 are the roots of the

$x^2-ax-a-1=0$, so

$m_1+m_2=a; m_1m_2=-(a+1)$

$\Rightarrow (m_1-m_2)^2=(m_1+m_2)^2+4m_1m_2$

$=a^2+4(a+1)=(a+2)^2$

so the required area is

$$\Delta=\pm\frac{a^2(a+2)}{2(a+1)}$$

since area is positive quantity, so we get two answers

35. Find the area of the triangle formed by the lines

$x+y=2$, $2x-2y=1$, $x=0$

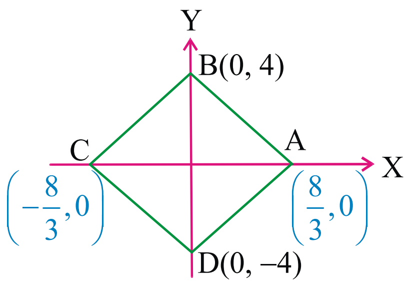
36. Let D is the mid point of AB

$OD=\frac{1}{\sqrt{2}}, AD=\sqrt{\frac{71}{2}}, AB=2AD$

37. required area $=\frac{p^2}{\sqrt{3}}$ where p is distance between parallel lines

38. Area of $\Delta^{le}=\frac{\Delta^2}{2\lambda_1\lambda_2\lambda_3}$ where Δ is determinent of 3 lines

39.



Enclosed Area ABCD= 4 Area AOB

$= 4\times\frac{1}{2}\times\frac{8}{3}\times 4 = \frac{64}{3}$ sq.units.

$\therefore f(n) = 2^n$

required area $= \frac{1}{3} f(6)$

40. $\frac{1}{4}\int_0^2 (x+k)\,dx = \frac{1}{2}(6)$.

41. From diagram $3\theta = 45^0 \Rightarrow \theta = 15^0$

42. from diagram $3\theta = 180^0 \Rightarrow \theta = 60^0$

43. Intersecting $2x+3y-5=0$ and $3x-5y+2=0$ is $(1,1)$

Find the image of $(1,1)$ w.r.to $x+y=\sqrt{2}$

44. If we reflect $y=|x-2|$ in y-axis, it will becomes $y=|-x-2|=|x+2|$. The reflected lines are $y=x+2,\ y=-x-2$. Their combined equation is

$(y-x-2)(y+x+2)=0$

$\Rightarrow y^2-(x+2)^2=0$

$\Rightarrow y^2-x^2-4x-4=0$

45. using symmetric form we get $A=(1,\sqrt{3})$, G devides $\overline{AD}$ in the ratio 2:1

46. $AC: 3x-2y=k$

$-3-6=k$, $3x-2y=-9$

$M:(1,6), \therefore C(3,9)\quad G(2,4)$

47. Triangle is accute angled

48. Triangle is obtuse angled

49. Let A(h,k), $D(\alpha,\beta)$ be the point on BC

Then

$\left(\frac{2\alpha+h}{3},\frac{2\beta+k}{3}\right)=(0,0)$ & $\alpha+\beta-2=0$

and $\left(\frac{k-0}{h-0}\right)(-1)=-1$

gives $\alpha=\beta=1$ and h=k=-2

Hence A(-2,-2)

50. $m_1=\frac{1}{2}, m=\frac{-2}{3}$, m_2 is slope of reflected line

use $\frac{m_1-m}{1+mm_1}=\frac{m-m_2}{1+mm_2}$

51. $ax+by=1$ will be one of the bisectors of the given line. Equation of bisectors of the given lines are

$\frac{3x+4y-5}{5}=\pm\left(\frac{5x-12y-10}{13}\right)$

$\Rightarrow 64x-8y=115$ or $14x+112y=15$

$\Rightarrow a=\frac{64}{115},\ b-\frac{8}{115}$ or

$a=\frac{14}{15},\ b=\frac{12}{115}$

52. $x_1=-10, y_1=2$ & $x_2=\frac{-19}{2}, y_1=\frac{3}{2}$

$x_3=\frac{-28}{3}, y_3=\frac{4}{3}$

53. Equation of AB is $x+y=21$

Number of integral solutions of $x+y<21$ is $20C_2$

54. Verification

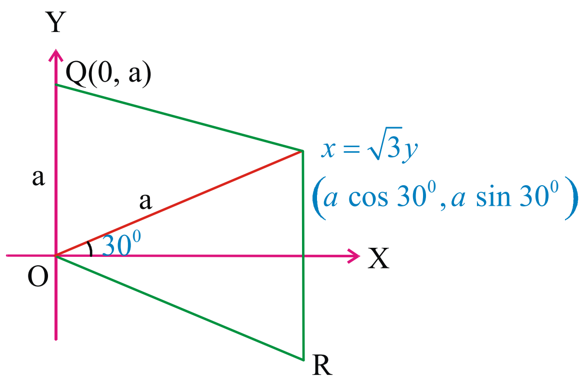
55. Verification

56. verification

57. Clearly Q, R are the two positions of the third vertex.

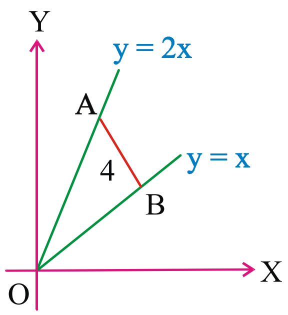
P(acos 30º, asin 30º), Q(0, a), R(a cos 30º, –a sin 30º)

i.e. $P\left(\frac{\sqrt{3}a}{2},\frac{a}{2}\right), Q(0,a)\ R\left(\frac{\sqrt{3}a}{2},\frac{-a}{2}\right)$



58. Let $B=(\alpha,\alpha)$ and middle point AB is (h,k)

Then, $A\equiv(2h-\alpha,2k-\alpha)$



lies on $y=2x$ then $\alpha=4h-2k.....(1)$

$|AB|=4$

$\Rightarrow[h-(4h-2k)]^2+[k-(4h-2k)]^2=4$

$\Rightarrow(-3h+2k)^2+(-4h+3k)^2=4$

or $25h^2+13k^2-36hk=4$

Required locus is $25x^2+13y^2-36xy-4=0$

Here, $h^2<ab$ and $\Delta\neq 0$

$\therefore$ ellipse

LEVEL-IV

1. **(A) : If** $(a_1x+b_1y+c_1)+(a_2x+b_2y+c_2)$ $+(a_3x+b_3y+c_3)$ **= 0, then the lines**

$a_1x+b_1y+c_1=0,$

$a_2x+b_2y+c_2=0, a_3x+b_3y+c_3=0$

can not be parallel

(R): If sum of three straight lines is identically 0 then they are either concurrent or parallel

1) A and R are true and R is the correct explaination of A
2) A and R are true and R not is the correct explaination of A
3) A is true R is False
4) A is False R is True

2. **A:** $(3,2)$ **is lies above the line** $x+y+1=0$

R: If the point $P(x_1,y_1)$ **lies above the line** $L=ax+by+c$ **then** $\dfrac{L(x_1,y_1)}{b}>0$

1) A and R are true and R is the correct explaination of A
2) A and R are true and R not is the correct explaination of A
3) A is true R is False
4) A is False R is True

3. **Assertion (A): If the angle between the lines kx-y+6 = 0, 3x+5y+7 = 0 is** $\pi/4$ **one value of k is -4**

Reason (R): If θ **is angle between the lines with slopes** m_1,m_2 **then** $\tan\theta=\left|\dfrac{m_1-m_2}{1+m_1m_2}\right|$.

1) A and R are true and R is the correct explaination of A
2) A and R are true and R not is the correct explaination of A
3) A is true R is False 4) A is False R is True

4. **I : Every first degree equation in x and y is ax+by+c=0,** $|a|+|b|\neq 0$ **represent a straight**

line

II : Every first degree equation in x and y can be convert into slope intecept form

Then which of the following is true

1) Only I 2) only II

3) both I & II 4) neither I nor II

5. **I : Length of the perpendicular from (x_1,y_1) to the line ax+by+c=0 is $\left|\frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}\right|$**

II : The equation of the line passing through(0,0) and perpendicular to ax+by+c=0 is bx-ay=0

Then which of the following is true.

1) onlyI 2)only II

3) both I & II 4) neither I nor II

6. **I : The ratio in which $L \equiv$ ax+by+c=0 divides the line segment joining $A(x_1,y_1)$ $B(x_2,y_2)$ is $\frac{-L_{11}}{L_{22}}$**

II: the equation of the line in which (x_1,y_1) divides the line segment between the coordinate axes in the ratio m:n is $\frac{nx}{x_1}+\frac{my}{y_1}=m+n$

Then which of the following is true

1) only I 2) only II

3) both I&II 4) neither I nor II

7. **I:A straight line is such that the algebraic sum of the distance from any no. of fixed points is zero. Then that line always passes through a fixed point**

II: The base of the triangle lie along the line x=a and is of length a. If the area of the triangle is a^2 then the third vertex lies on x=-a or x=3a.

Then which of the following is true.

1) only I 2) only II

3) both I & II 4) neither I nor II

8. **<u>Statement I:</u> Normal form of line $x+y=\sqrt{2}$ is $x\cos\frac{\pi}{4}+y\sin\frac{\pi}{4}=1$**

<u>Statement II:</u> The ratio in which the perpendicular through (4,1) devides the line joining (2,-1), (6,5) is 5:8

Which of the above statement (s) is/are true

1) Only I 2) Only II

3) Both I and II 4) Neither I nor II

9. **The lines $L_1 : y-x=0$ and $L_2 : 2x+y=0$ intersect the line $L_3 : y+2=0$ at P and Q respectively. The bisector of the accute angle between L_1 and L_2 intersect L_3 at R**

Statement - 1 : The ratio PR : RQ equals $2\sqrt{2}:\sqrt{5}$.

Statement - 2 : In any triangle, bisector of an angle divides the triangle into two similar triangles. [AIEEE - 2011]

(1) Statement – 1 is true, Statement – 2 is true; Statement – 2 is not a correct explanation for Statement – 1

(2) Statement – 1 is true, Statement– 2 is false.

(3) Statement – 1 is false, Statement– 2 is true.

(4) Statement – 1 is true, Statement – 2 is true; Statement – 2 is a correct explanation for Statement – 1

10. **Observe the following list with respect to the line ax+by+c=0**

List I	**List II**
A)Perpendicular distance from (0,0)	1) $^{-c}/_{b}$
B) X-intercept of the line	2) $\left(\frac{-c}{a},\frac{-c}{b}\right)$
C) Y-intercept of the line	3) $\frac{\lvert c\rvert}{\sqrt{\lvert a^2+b^2\rvert}}$
D) Circumcentre of triangle OAB	4) $\left[\frac{-c}{2a},\frac{-c}{2b}\right]$
where A,B are X and Y intercepts	5) $^{-c}/_{a}$

Then the correct answer is

	A	B	C	D		A	B	C	D
1)	3	5	1	2	2)	3	5	1	4
3)	3	4	1	5	4)	1	2	3	4

11. **observe the following**

column I	**column II**
A) the area bounded by the curve max $\{\lvert x\rvert,\lvert y\rvert\}=1$ is	**P) 3**
B) if the point (a,a) lies between	**Q) 2**

the lines $|x+y|=3$ then number

of values of $[|a|]$

is (where [.] denotes the greatesr integer function)

C) Number of integral values of b for which the origin and the point(1,1) lie on the same side of the st.line **R) 4**

$a^2x+aby+1=0$ **for all** $a \in R \sim \{0\}$

1) $A \to R, B \to Q, C \to P$

2) $A \to R, B \to P, C \to Q$

3) $A \to P, B \to Q, C \to R$

4) $A \to P, B \to R, C \to Q$

12. **Equation of line passing through (1,3), perpendicular to 2x-3y+4 = 0 is ax+by+c = 0 (a>0) then ascending order of a, b, c is**

1) a, c, b 2) c, b, a
3) c, a, b 4) a, b, c

LEVEL-IV - KEY

1) 4	**2) 1**	**3) 1**	**4) 1**	**5) 3**
6) 3	**7) 3**	**8) 3**	**9) 2**	**10) 2**
11) 1	**12) 2**			

LEVEL-IV - HINTS

1. Given lines may be parallel.
2. R is the correct explaination of A.
3. Using angle formula between the lines
4. Standard result
5. Standerd result
6. Standerd result
7. I: Consider the points

$(x_1,y_1)(x_2,y_2)..........(x_n,y_n)$

Alegebric sum of the perpendicular distance =

to $\dfrac{a(x_1+x_2..........+x_n)+b(y_1+y_2........+y_n)+nc}{\sqrt{a^2+b^2}}$

II : Consider the points on the line x = a are A(a,0) B(a,a)

P(x,y) area of the triangle = a^2

8. I) normal form line $x\cos\alpha + y\sin\alpha = p$

II) $-L_{11} : L_{22}$

9. $P(2,-2), Q(1,-2)$

Equation of angular bisector OR is

$(\sqrt{5}+2\sqrt{2})x = (\sqrt{5}-\sqrt{2})y$

$\therefore PR : RQ = 2\sqrt{2} : \sqrt{5}$

10. A) the perpendicular distance from $(0,0)$ to

$ax+by+c=0$ is $\dfrac{|c|}{\sqrt{a^2+b^2}}$

B) X-Intercept $\dfrac{-c}{a}$

C) Y-intercept $\dfrac{-c}{b}$

D) In right angle triangle circumcenter is mid point of hypotenuse is

11. (A) $\because$ max $\{|x|,|y|\}=1$

If $|x|=1$ and if $|y|=1$,

then $x=\pm1$ and $y=\pm1$

$\therefore$ Required area $=2\times2$

(B) the line $y=x$ cuts the lines $|x+y|=3$ ie,

at or and

then ,

(C) since (0, 0) and (1, 1) lie on the same side.

So,

Coefficient of is >0

or

Number of values of is 3.

12. Line equation is 3x+2y-9 = 0.

****